MUHS final-year BPTh (IV BPTh), 2012 syllabus

Neuro Physiotherapy

Model answers to every 3-mark and 7-mark question asked from Summer 2016 to Summer 2026, written the way to write them in the exam.



Questions	All 237 short-answer slots of the 20 papers (code 51412): 79 distinct 3-mark and 73 distinct 7-mark questions after merging repeats
Arranged	By topic, in syllabus order, with an index by paper at the end
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/neuro/

Educational notes for exam preparation. Not medical advice. Every answer is drawn from the full notes, which list the sources for each fact.

How to use the model answers

Every short-answer question from the 20 MUHS papers (Summer 2016 to Summer 2026) is here, once. When a question was repeated, the sessions are listed under it, so you can see which ones come back.

1. **Read the question, close the book, write your own answer** in the time allowed, then compare. Copying without recall does not stick.
2. **Learn the structure** as much as the facts: definition, points, table or diagram, physiotherapy, one closing line.
3. **Practise whole papers.** The index at the back lists each paper's short questions with the page of every answer: sit one paper in 100 minutes, then mark it.

What examiners look for

Question	Length and time	Structure
3 marks (any 5 of 6)	Half a page; about 6 minutes	One-line definition, then 3–5 points, a small table or a labelled sketch; one line of physiotherapy relevance
7 marks (any 5 of 6)	One and a half to two pages; about 14 minutes	Definition or introduction; classification or features; 6–8 headed points or a table with doses; a labelled diagram; precautions; one closing line

Write **doses with units** and **underline the keywords**. A **"Draw"** line tells you which diagram to draw; where the full notes have it, the diagram is printed beside the answer.

> **The paper:** 80 marks in 3 hours. Q1: any 5 of 6 at 3 marks (15). Q2: any 5 of 6 at 7 marks (35). Q3 and Q4: one 15-mark long answer each (30). About 2 minutes a mark. The long answers are covered in the full notes.

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1 Application of the ICF model in neurological conditions

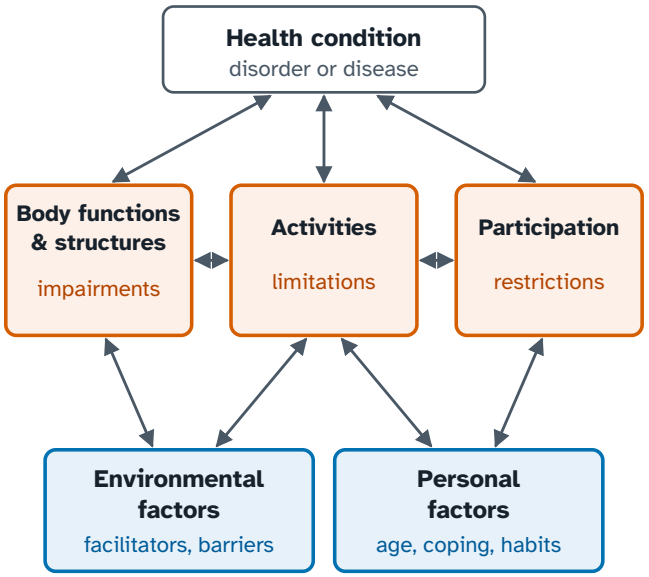
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7 marks Explain domains of ICF

Asked: S19 (2d)

ICF (International Classification of Functioning, Disability and Health, **WHO 2001**) is a **biopsychosocial** framework: disability is the result of an interaction between a health condition and contextual (environmental and personal) factors. **Functioning** is the positive side, **disability** the negative side.

The ICF model



Draw this first in every ICF answer. The arrows go both ways: every box can change every other box.

Component (code)	Domains (chapters)	Neuro example
Body functions (b)	Mental; sensory and pain; voice and speech; cardiovascular, haematological, immunological, respiratory; digestive, metabolic, endocrine; genitourinary, reproductive; **neuromusculoskeletal and movement-related**; skin	Muscle power b730, tone b735
Body structures (s)	Nervous system; eye and ear; voice and speech; cardiovascular and respiratory; digestive; genitourinary; structures related to movement; skin	Brain, spinal cord, nerve
Activities and participation (d)	Learning and applying knowledge; general tasks; communication; **mobility**; self-care; domestic life; relationships; major life areas (education, work); community, social and civic life	Transferring d420, walking d450
Environmental factors (e)	Products and technology; natural and built environment; support and relationships; attitudes; services, systems and policies	Walking aid e120, family e310
Personal factors	Not classified: age, sex, education, coping, habits, motivation	Motivated young teacher

Qualifiers: 0 no, 1 mild, 2 moderate, 3 severe, 4 complete problem (b730.3 = severe weakness); environmental factors take + for facilitators and – (or plain number) for barriers.

Uses: common language for the team, functional diagnosis, SMART goals at each level, capacity vs performance, documentation for disability certificates. The arrows are two-way: treat at more than one level.

2 Motor control: theories behind neurological physiotherapy

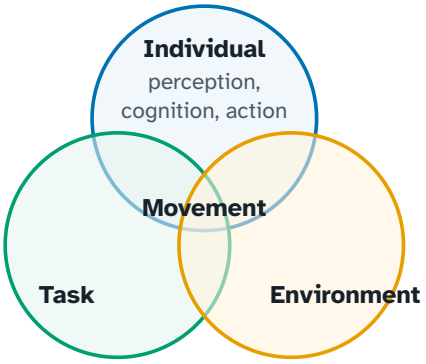
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7 marks Explain theories of motor control (any two)

Asked: W19 (2c), S21 (2b), S23 (2d) (asked 3 times)

Motor control is the ability to regulate or direct the mechanisms essential to movement. Movement emerges from the **individual**, the **task** and the **environment**; the theory a therapist follows decides what is treated.

Movement emerges from the individual, the task and the environment



The systems view: change any of the three, and the movement changes. Treatment therefore works on the person, the task and the setting.

1. Hierarchical theory (Hughlings Jackson, Magnus)

- Control is **top-down**: cortex → brainstem → spinal cord; development follows a fixed sequence.
- A higher lesion **releases primitive reflexes** and abnormal patterns (for example after stroke).
- **Limitation:** lower levels also influence higher ones; not all movement is hierarchical.
- **Use:** basis of neurofacilitation approaches (Bobath, Brunnstrom, Rood): inhibit reflexes, follow developmental sequences.

2. Systems theory (Bernstein)

- The body is a mechanical system with many **degrees of freedom**; the nervous system controls them through **synergies**.
- Movement arises from many systems working together (musculoskeletal, sensory, perceptual, cognitive) plus forces from body and environment.
- **Limitation:** less detail on how nervous system and environment interact.
- **Use:** assess and treat every system that limits the task (strength, range, sensation, balance); basis of **task-oriented training**.

Other theories in one line each: **reflex** (Sherrington: movement as chains of reflexes; cannot explain movement without input); **motor programming** (central pattern generators; treadmill stepping); **dynamic systems** (self-organisation, attractor states, control parameters such as speed); **ecological** (Gibson: perception–action, affordances; practise real tasks in real places).

> **Draw:** The individual, task and environment circles; cortex over brainstem over spinal cord.

Part 3: Plasticity and recovery, strength training and skill acquisition

3 Neuroplasticity, recovery, motor learning and skill acquisition

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3 marks Neuroplasticity: definition, brain changes and synaptogenesis

Asked: S16 (1a), S18 (1f), S26 (1c) (asked 3 times)

Neuroplasticity is the ability of the nervous system to change its **structure and function** in response to experience, learning, training or injury. It can be adaptive (recovery) or maladaptive (spasticity, learned non-use).

- **Synapse:** stronger or weaker transmission (long-term potentiation, depression); unmasking of silent synapses.
- **Neuron: synaptogenesis** (new synapses), new dendritic spines, **axonal sprouting** from intact neurons into areas that lost input.
- **Cortical maps:** trained areas enlarge; areas next to a lesion take over lost functions.

Synaptogenesis is the formation of new synapses: fastest in early development, continues with learning, and after injury intact neurons sprout new synapses (reactive synaptogenesis) that practice can shape. Principles: use it or lose it, specificity, repetition, intensity, salience (Kleim and Jones, 2008).

3 marks Three factors that facilitate motor learning

Asked: W17 (1a)

1. **Practice:** many repetitions; distributed rather than massed; **variable and random** practice for retention and transfer; whole-task practice where possible.
2. **Feedback:** the right type and amount (knowledge of results and of performance), more early and **faded** later to avoid dependence.
3. **Motivation and attention:** meaningful, task-specific goals in a real-life context; an **external focus** ("reach the cup") rather than on the body part.

Also: demonstration and mental practice, sleep and rest, a stimulating environment.

3 marks Motor learning and its stages

Asked: S21 (1e), W24 (1e), W25 (1e) (asked 3 times)

Motor learning is a set of processes linked to practice or experience that leads to **relatively permanent** changes in the capability for skilled movement, shown by retention and transfer.

Stage (Fitts and Posner)	Learner	Therapist
Cognitive	Works out what to do; slow, many errors, full attention	Clear instructions, demonstration, frequent feedback, quiet setting
Associative	Refines how to do it; fewer errors, more consistent	Less feedback, more practice, start varying the task
Autonomous	Automatic; can do two tasks at once	Varied real-world practice, dual tasks, little feedback

3 marks Extrinsic and intrinsic feedback

Asked: S22 (1a)

- **Intrinsic feedback:** information from the person's own senses during and after movement: proprioception, vision, vestibular, touch (feeling the heel strike, seeing the cup move).
- **Extrinsic (augmented) feedback:** added from outside to supplement intrinsic feedback: therapist's words, mirror, video, EMG biofeedback, force platforms. It is **knowledge of results** (outcome: "10 m in 12 s") or **knowledge of performance** (pattern: "your knee locked back").

In neurological patients extrinsic feedback replaces lost intrinsic feedback (vision for poor proprioception) and is **faded** as learning progresses.

7 marks Role and types of feedback in neuro-rehabilitation

Asked: S25 (2b)

Feedback is information about a movement that the learner uses to correct errors and learn the skill.

Type	Meaning and example
Intrinsic	From own senses: feeling the foot hit the floor, seeing the target
Extrinsic (augmented)	From outside: verbal cues, mirror, video, EMG biofeedback, force platform, virtual reality
Knowledge of results (KR)	About the outcome: "you stood up in 5 seconds"
Knowledge of performance (KP)	About the pattern: "keep your weight over the weak leg"
Concurrent or terminal	During the movement (mirror) or after it (verbal)
Immediate or delayed; summary, faded, bandwidth	Schedules that reduce dependence: feedback after several trials, less over time, or only when error exceeds a band

Role

- **Information** to detect and correct errors.
- **Motivation** and **reinforcement** of correct movement.
- **Substitution** for lost intrinsic feedback: vision for proprioceptive loss, EMG biofeedback for a flicker of activity, mirror therapy.

Rules

- More feedback in the cognitive stage; **fade** it so the patient relies on intrinsic feedback (too much creates dependence and poor retention).
- Give KP early, KR later; keep it short with an external focus.

Example: sit-to-stand after stroke: mirror and "nose over toes" (KP) early; number of stands in 30 seconds (KR) later.

Part 4: Quality of life scales and independence measures

4 Quality of life, independence and other outcome measures

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3 marks Glasgow Coma Scale: components, interpretation and uses

Asked: W16 (1c), S19 (1a), S20 (1b), S26 (1d) (asked 4 times)

Eye opening (E 1–4)	4 spontaneous, 3 to sound, 2 to pressure, 1 none
Verbal (V 1–5)	5 oriented, 4 confused, 3 words, 2 sounds, 1 none
Motor (M 1–6)	6 obeys, 5 localises, 4 normal flexion, 3 abnormal flexion, 2 extension, 1 none

Total 3–15: severe **3–8** (coma; airway at risk), moderate 9–12, mild 13–15. Report the parts (E2 V2 M4 = 8); the **motor score** predicts outcome best; "T" if intubated.

Used in: head injury, stroke, meningitis and encephalitis, raised ICP, overdose; serial scores show deterioration.

3 marks Modified Ashworth Scale

Asked: S19 (1e), W21 (1f) (asked 2 times)

Grades **resistance to passive movement** (spasticity) with the patient relaxed, moving the part through full range in about one second.

0	No increase in tone
1	Slight increase: catch and release, or minimal resistance at the end of range
1+	Slight increase: catch, then minimal resistance through less than half the remaining range
2	More marked increase through most of the range, part still moves easily
3	Considerable increase; passive movement difficult
4	Rigid in flexion or extension

Quick and clinical, but does not separate neural from stiffness components (the Tardieu scale does).

3 marks ASIA scale

Asked: S19 (1c)

The **ASIA Impairment Scale** grades completeness of spinal cord injury from the ISNCSCI examination (key muscles and dermatomes, plus **S4–S5**).

- **A, complete:** no sensory or motor function in S4–S5.
- **B, sensory incomplete:** sensation but no motor function below the level, including S4–S5.
- **C, motor incomplete:** motor function below the level, but less than half the key muscles below the neurological level are grade 3 or more.
- **D, motor incomplete:** half or more of those key muscles are grade 3 or more.
- **E, normal:** normal sensation and power in a person who had deficits.

Use: classification, prognosis (incomplete injuries recover more), goals.

3 marks Hoehn and Yahr scale

Asked: W20 (1a)

Stages the severity of **Parkinson's disease** by spread of signs and balance.

1. **Stage 1:** one side only.
2. **Stage 2:** both sides, balance normal.
3. **Stage 3:** both sides, some postural instability; physically independent.
4. **Stage 4:** severe disability; still walks or stands without help.
5. **Stage 5:** wheelchair-bound or bedridden unless helped.

The modified scale adds 1.5 (one side plus trunk) and 2.5 (both sides, recovers on the pull test). Use: describe stage, guide goals (balance and falls work from stage 3).

Part G: Assessment for every condition

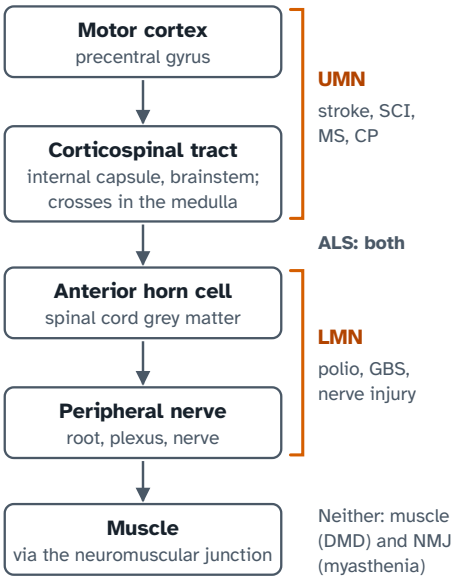
G Neurological assessment and investigations

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3 marks UMN and LMN lesions: features and differences

Asked: S16 (1d), W16 (1a), S20 (1c), W21 (1e), W22 (1d), S24 (1f) (asked 6 times)

Where UMN and LMN lesions lie



Draw this beside a UMN vs LMN answer. The anterior horn cell is where the two meet: the UMN ends there and the LMN starts there.

Feature	UMN lesion	LMN lesion
Site	Cortex, internal capsule, brainstem, spinal cord tracts	Anterior horn cell, root, plexus, nerve
Tone	**Spasticity** (clasp-knife)	**Flaccidity**
Reflexes	**Exaggerated**, clonus	Reduced or absent
Plantar	**Extensor (Babinski)**	Flexor or absent
Wasting	Little (disuse only)	**Marked**, early
Fasciculation	Absent	May be present
Weakness	Groups and patterns (pyramidal distribution), synergies	Individual muscles in a root or nerve distribution

Examples: stroke, SCI (UMN); nerve injury, GBS, polio (LMN). MND shows both.

G Cognition, perception, apraxia and aphasia

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3 marks Figure-ground discrimination

Asked: W18 (1a)

The inability to **pick out an object from its background** (a visual perceptual disorder, usually after right parieto-occipital damage).

- **Examples:** cannot find a white shirt on a white bed, the brake on a wheelchair, or a step edge on a patterned floor.
- **Assessment:** find an item in a cluttered drawer or picture; Ayres' figure-ground test; observe daily tasks.
- **Physiotherapy:** reduce clutter; use **colour contrast** (tape on step edges and brakes); teach a systematic search; practise in real settings.

3 marks Combined (cortical) sensations

Asked: W18 (1f)

Sensations that need intact primary sensation plus **integration in the parietal cortex**; tested with the eyes closed.

- **Stereognosis:** recognising an object (coin, key) by touch.
- **Graphaesthesia:** recognising a number drawn on the skin.
- **Two-point discrimination:** telling one point from two (fingertip about 2–5 mm).
- **Tactile localisation; barognosis** (weight); texture recognition.
- **Bilateral simultaneous stimulation:** one side "disappears" in extinction.

Loss with intact primary sensation points to a parietal lesion (stroke); it limits hand use, so add sensory re-education.

3 marks Assessment of unilateral visual neglect

Asked: S19 (1d)

Neglect: failure to report, respond or orient to stimuli on the side opposite a brain lesion, not due to weakness or sensory loss; commonest after **right parietal** lesions (left neglect).

- **Line bisection:** mark is shifted towards the good side.
- **Cancellation tests** (star, letter, Albert's lines): targets missed on the neglected side.
- **Clock drawing, figure copying:** one half missing.
- **Reading and writing:** starts mid-line, drifts to one side.
- **Functional observation:** eats half the plate, bumps door frames (Catherine Bergego Scale).

Tell it from hemianopia: a hemianopic patient knows of the loss and turns the head to compensate; a neglect patient does not.

3 marks Apraxia and its types

Asked: S22 (1e)

Apraxia: inability to carry out learned purposeful movements on command despite adequate strength, sensation, coordination, comprehension and willingness (left parietal or frontal lesion).

- **Ideomotor:** cannot do it on command or by imitation, but does it automatically.

- **Ideational:** loses the idea or sequence of a task (toothpaste on the comb).
- **Limb-kinetic:** clumsy fine finger movements.
- **Constructional, dressing, buccofacial, gait** apraxia.

3 marks Aphasia and its types; Broca's and Wernicke's aphasia

Asked: S23 (1f), S25 (1c) (asked 2 times)

Aphasia: loss of language (speaking, understanding, reading, writing) from brain damage, usually left hemisphere; unlike **dysarthria**, which is a motor speech problem with normal language.

Type	Features	Lesion
Broca's (expressive, non-fluent)	Slow, effortful, telegraphic speech; comprehension fairly good; aware and frustrated	Left inferior frontal gyrus
Wernicke's (receptive, fluent)	Fluent but meaningless speech with wrong and made-up words; poor comprehension; often unaware	Left posterior superior temporal gyrus
Conduction	Fluent, comprehension good, repetition very poor	Arcuate fasciculus
Global	All language functions severely lost	Large left MCA lesion
Anomic; transcortical	Word-finding problems; repetition preserved in transcortical types	Various

PT: short sentences, gestures, demonstration, yes/no questions; check understanding by the action.

7 marks Apraxia and its types, with examples

Asked: S23 (2f), S24 (2d), W24 (2f) (asked 3 times)

Apraxia is the inability to carry out learned, purposeful movements on command, even though strength, sensation, coordination, comprehension and willingness are adequate. It usually follows a **left parietal or frontal** lesion (stroke, head injury, dementia).

Type	What fails	Example
Ideomotor	Movement on command or imitation; can do it automatically (left parietal)	Cannot show how to comb when asked, but combs in the morning
Ideational (conceptual)	The idea and sequence of a task; misuse of objects	Toothpaste on the comb; drinks tea before pouring it
Limb-kinetic	Precise fine finger movements (premotor cortex)	Cannot button a shirt without weakness
Constructional	Copying, drawing or building shapes (parietal)	Cannot copy a cube or block design
Dressing	Dressing correctly (often right parietal, with neglect)	Shirt inside out or on one arm
Buccofacial (oral)	Facial movements on command (left frontal, with Broca's)	Cannot blow out a candle when asked
Gait	Starting and organising walking (frontal, hydrocephalus)	Feet "glued to the floor"

Assessment

Pantomime tool use ("show me how you brush your teeth"), imitation of gestures, real-object use, a multistep task (making tea); note wrong movement, body part as tool, wrong order, perseveration.

Physiotherapy

- Practise **real tasks with real objects** in their natural setting.
- **Strategy training:** talk through each step, picture or step cards.
- **Errorless learning and chaining** (forward or backward); physical guidance rather than long verbal commands; fade cues.
- Gesture training for ideomotor apraxia; involve the family.

Part 5A: Physiotherapy management: adult conditions

5A(a) Stroke

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3 marks Types of stroke

Asked: S26 (1f)

Stroke (WHO): rapidly developing signs of focal disturbance of brain function lasting **more than 24 hours** or leading to death, with no cause other than vascular.

Type	Mechanism	Clue
Ischaemic: thrombotic	Clot on an atherosclerotic plaque	Onset at rest or in sleep; stepwise; earlier TIAs
Ischaemic: embolic	Clot from the heart (AF, rheumatic valve) or carotid	Sudden, maximal at onset
Ischaemic: lacunar	Small deep artery blocked	Pure motor or pure sensory; no cortical signs
Haemorrhagic: intracerebral	Small artery ruptures (hypertension)	Headache, vomiting, falling consciousness
Haemorrhagic: subarachnoid	Aneurysm bursts	"Worst headache of my life", neck stiffness

By time course: **TIA**, stroke in evolution, completed stroke. About 85% are ischaemic in Western series; haemorrhage is commoner in India (a fifth to a third).

3 marks Anterior cerebral artery syndrome

Asked: S26 (1a)

The ACA supplies the medial surface of the frontal and parietal lobes, including the **leg area** of the motor and sensory cortex. Occlusion gives, on the **opposite side**:

- Weakness and sensory loss of the **leg more than the arm** (face spared).
- **Urinary incontinence**.
- **Abulia** (apathy, slowed thinking), personality change; **grasp reflex**.
- Difficulty **starting gait** (gait apraxia).

Contrast: MCA gives face and arm more than leg, with aphasia or neglect.

3 marks Flexion synergy of the spastic upper limb

Asked: S18 (1b)

A **synergy** is a fixed, stereotyped pattern in which muscles contract together and cannot act alone, seen in Brunnstrom stages 2–3 after stroke.

Part	Component
Scapula	Retraction and elevation
Shoulder	Abduction and external rotation
Elbow	**Flexion** (the strongest component)
Forearm	Supination
Wrist and fingers	Flexion

The typical posture mixes the strongest parts of both synergies: elbow flexion (flexion synergy) with shoulder adduction and internal rotation (extension synergy).

3 marks Pusher syndrome

Asked: S20 (1d), W24 (1b) (asked 2 times)

- **Definition:** the patient **pushes** with the non-paretic arm and leg **towards the paretic side** and **resists correction**, so falls towards the weak side in sitting and standing.
- **Cause:** a disturbed sense of upright body posture, often after **right-hemisphere** or **thalamic** lesions; graded with the Scale for Contraversive Pushing.
- **Treatment:** use vision to show what is vertical (mirror, door frame, therapist's arm); reach towards the **strong side**; sit or stand with the strong side against a wall or high table to remove the pushing surface. Do not force the patient upright.

3 marks Three techniques for the spastic hand

Asked: S17 (1b)

1. **Slow sustained stretch and weight-bearing** through the open hand (wrist and fingers extended) in sitting.
2. **Prolonged icing** or slow rhythmic rotation to reduce tone, followed at once by active grasp and release practice.
3. **Electrical stimulation** of wrist and finger extensors (or mirror therapy, CIMT when eligible).

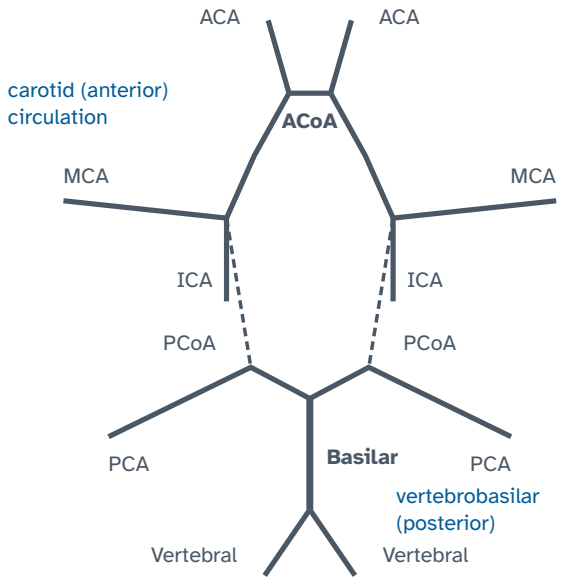
Botulinum toxin plus therapy for marked focal spasticity.

7 marks Circle of Willis with a diagram

Asked: S20 (2b)

The **circle of Willis** is a ring of arteries at the base of the brain, around the optic chiasma and pituitary, joining the two internal carotid systems and the vertebrobasilar system.

The circle of Willis, seen from below



A ring of arteries at the base of the brain. The communicating arteries (dashed) let blood cross over if one feeding artery narrows slowly. ACA anterior cerebral, ACoA anterior communicating, ICA internal carotid, MCA middle cerebral, PCoA posterior communicating, PCA posterior cerebral.

Vessels

- **Front:** right and left **anterior cerebral arteries** (ACA) joined by the **anterior communicating artery** (ACoA).
- **Sides:** each **internal carotid** (ICA) divides into the ACA and the **middle cerebral artery** (MCA); the **posterior communicating artery** (PCoA) joins the ICA to the PCA.
- **Back:** two vertebral arteries form the **basilar artery**, which divides into the two **posterior cerebral arteries** (PCA).

Supply

- Carotid (anterior) circulation: most of the cerebral hemispheres (ACA medial surface and leg area; MCA lateral surface, face and arm area, speech areas).
- Vertebrobasilar (posterior) circulation: brainstem, cerebellum, occipital lobes, thalamus.

Function and clinical importance

- A **safety connection** (collateral) between right and left and between front and back: if one artery narrows slowly, blood crosses over.
- Often incomplete or variable, so protection differs between people.
- Most **berry aneurysms** causing subarachnoid haemorrhage arise at its junctions, especially the ACoA.
- Occlusion of each branch gives a typical syndrome: MCA (face and arm, aphasia or neglect), ACA (leg, incontinence), PCA (hemianopia, memory).

Draw: The ring from below: ACAs and ACoA at the top, ICAs at the sides giving off MCAs, PCoAs dashed, basilar dividing into PCAs at the bottom.

7 marks Physiotherapy in the acute stage of CVA

Asked: S17 (2a)

Aims: prevent complications (chest infection, pressure injury, DVT, shoulder pain, contracture), start early functional movement, educate the family.

Timing: begin once medically stable, with short, frequent sessions. AVERT (2015): very early, high-dose out-of-bed activity in the first 24 hours worsened outcome.

Problem	Physiotherapy
Positioning	Change every 2 hours: on the affected side (shoulder brought forward), on the unaffected side (affected arm forward on a pillow), supine (pillow under affected shoulder and hip); supported sitting early; head up 30° with swallowing or chest risk
Chest	Deep breathing, assisted cough, positioning; watch for aspiration; nothing by mouth until a swallow screen is passed
Shoulder	Support the arm in sitting; never lift or pull by the arm
Range	Gentle passive or assisted ROM of all affected joints daily, pain-free
Function	Rolling to both sides, lying to sitting, sitting balance, sit to stand, standing, short assisted walks
DVT	Active movements, early mobility, intermittent pneumatic compression if immobile
Stimulation	Approach and place objects on the affected side; sensory input to the limbs

Precautions: follow the unit's BP limits (after thrombolysis or haemorrhage); no pressure on the craniectomy side; stop for new weakness, headache, falling consciousness, chest pain or seizure.

Teach the family positioning and safe handling before discharge.

7 marks Physiotherapy of the hemiplegic shoulder

Asked: S16 (2d), S26 (2f) (asked 2 times)

Problems

- **Inferior subluxation** in the flaccid stage: the weight of the arm stretches the capsule (gap below the acromion, measured in finger-breadths).
- **Pain** (impingement, poor handling, spasticity).
- **Spasticity** of internal rotators and adductors; loss of external rotation.
- **Shoulder-hand syndrome** (CRPS): painful shoulder with a swollen, discoloured hand.

Prevention

- Support the arm on a pillow, **arm trough** or lap tray in sitting; correct positioning in bed.
- Handle with the **scapula moved forward**; **never pull on the arm** in transfers; teach staff and family.

Treatment

- ROM within the pain-free range, **scapula mobilised first** (protraction, upward rotation) before elevating the arm.
- Weight-bearing through the arm in sitting; active-assisted then task practice as movement returns.
- **Electrical stimulation** of supraspinatus and posterior deltoid for subluxation.
- **Botulinum toxin** for a spastic, painful shoulder, followed by stretching.
- Shoulder-hand syndrome: elevation, gentle active movement, desensitisation; refer early.

Avoid overhead pulleys: the 2026 AHA/ASA guideline rates them harmful because they increase shoulder pain.

7 marks Hemiplegic (spastic) gait: features and physiotherapy

Asked: S18 (2d), S19 (2f), S21 (2f), S22 (2b), S25 (2c) (asked 5 times)

Phase	What goes wrong	Main reason
Stance	Short stance on the weak leg; knee hyperextends or stays bent; trunk leans; little hip extension at the end of stance; weak push-off	Weak hip extensors and quadriceps; spastic or tight calf; poor balance
Swing	Stiff knee; foot drop and inversion; **circumduction** or hip hiking	Weak hip flexors and dorsiflexors; spastic quadriceps and calf (**extensor synergy**)
Overall	Slow, short, asymmetric steps; arm held flexed; high energy cost	Weakness, spasticity, poor selective control, fear

Physiotherapy

1. **Treat the cause:** strengthen hip extensors, quadriceps and dorsiflexors; stretch the calf; reduce tone before practice.
2. **Practise the parts:** weight shift onto the weak leg, step-ups, stepping forwards and backwards, knee control in small ranges, hip extension in late stance.
3. **Practise the whole:** repetitive overground walking and treadmill training (with or without body-weight support), faster as able.
4. **Foot drop:** AFO or peroneal nerve stimulation; walking aid only as needed.
5. **Progress:** outdoors, slopes, obstacles, dual tasks; measure with the 10-metre walk and 6-minute walk.

Through the phases of recovery

- **Flaccid:** standing and stepping in parallel bars with knee support.
- **Synergy:** walking dominated by extensor synergy, so break it: knee bending with the hip extended, dorsiflexion with the knee straight.
- **Selective control:** faster, more symmetrical gait; training moves to the community.

7 marks Physiotherapy of the spastic hand

Asked: W16 (2d), S24 (2a) (asked 2 times)

After stroke the wrist and finger flexors become spastic, so the hand stays fisted; grasp is possible but **release** is not, and contracture and hygiene problems follow.

Assess: MAS of wrist and finger flexors, passive range, active extension, sensation, Brunnstrom hand stage.

- **Stretch and position:** slow, sustained stretch of wrist and finger flexors; **weight-bearing through the open hand** in sitting.
- **Reduce tone before practice:** slow rhythmic rotation of trunk and shoulder; **prolonged icing**; reflex-inhibiting positions.
- **Active practice:** grasp and release of objects of different sizes; task practice with many repetitions.
- **Electrical stimulation** of wrist and finger extensors; **mirror therapy**; **CIMT** if at least 20° active wrist and 10° finger extension.
- **Botulinum toxin** to finger and wrist flexors, followed by stretching and task practice.

- **Splints:** resting splints do not prevent spasticity (AHA/ASA 2026); use only for a specific reason (developing contracture, hygiene) and check the skin.

Home programme: daily self-stretch with the strong hand, weight-bearing on the open hand, and using the hand as a helper.

7 marks Physiotherapy for the flaccid upper limb after stroke

Asked: S22 (2a)

In the flaccid stage (Brunnstrom 1) there is no tone and no voluntary movement; the risks are **shoulder subluxation**, pain, oedema and **learned non-use**.

- **Protect the shoulder:** arm trough or lap tray in sitting, positioning in bed, handle with the scapula forward, never pull on the arm.
- **Facilitate activity (Rood):** quick stretch, tapping, fast brushing and vibration over the muscles to be activated (triceps, wrist extensors, deltoid).
- **Weight-bearing** through the extended arm in sitting (approximation).
- **Placing and holding** the arm in different positions.
- **Electrical stimulation** of supraspinatus, deltoid and wrist extensors.
- **Mirror therapy** and **bilateral arm training**.
- Passive then **active-assisted**, then active task practice as the first movements return; keep full pain-free range.
- Oedema: elevation, active movement of the hand.

EPOS: some finger extension and shoulder abduction within 72 hours predicts arm recovery.

7 marks Motor Relearning Programme for the hemiplegic upper limb

Asked: W18 (2c)

Carr and Shepherd: a task-oriented approach based on motor learning. The patient is an active learner practising **real tasks**; muscle activity is trained in its functional context.

Four steps

1. **Analyse the task:** observe reach, grasp and manipulation; find the **missing components** and compensations.
2. **Practise the missing components:** forward reach with elbow extension; wrist extension while grasping; opening the hand to release; thumb opposition; supination and pronation.
3. **Practise the whole task:** reach, grasp, move and release a glass; pick up and place objects.
4. **Transfer** into daily life with practice outside therapy time (written programme, family supervision).

Principles

- Clear goal, instruction and demonstration; feedback faded over time.
- Many repetitions; avoid unnecessary muscle activity; do not let the strong hand take over.
- Prevent soft-tissue shortening and learned non-use.

5A(b) Acquired brain injury: trauma and space-occupying lesions

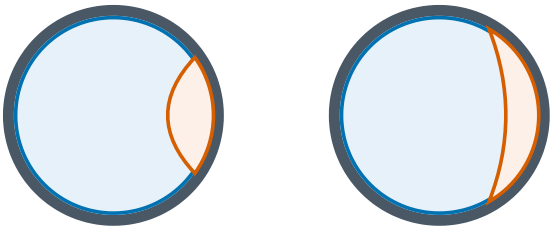
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3 marks Classification of head injury

Asked: S24 (1a)

Basis	Types
Severity (GCS)	**Mild 13–15**; **moderate 9–12**; **severe 3–8**
Mechanism	Closed (blunt) or open (penetrating); acceleration–deceleration; blast
Skull	Linear, depressed or basal fracture (CSF leak, raccoon eyes)
Focal lesions	Extradural haematoma, subdural haematoma, intracerebral haemorrhage, contusion
Diffuse lesions	Concussion, diffuse axonal injury, diffuse swelling
Timing	**Primary** (at impact); **secondary** (later: hypoxia, hypotension, raised ICP, seizures, infection)

Extradural vs subdural haematoma on a CT slice



Extradural
lens-shaped, lucid interval

Subdural
thin crescent

Extradural: an arterial bleed (usually the middle meningeal artery after a temporal fracture) that strips the dura and stops at the sutures, so it is lens-shaped; the person may talk normally before collapsing (lucid interval). Subdural: torn bridging veins; the blood spreads along the brain surface as a crescent; acute after major trauma, chronic in older people after a minor fall.

3 marks Features (signs) of raised intracranial pressure

Asked: W22 (1f), S25 (1d) (asked 2 times)

- **Symptoms:** headache worse on waking, coughing and bending; vomiting (may be projectile); blurred vision.
- **Signs:** falling level of consciousness (GCS); **papilloedema**; a **dilating pupil** on the side of the mass (third nerve); sixth nerve palsy; seizures; bulging fontanelle in infants.

- **Cushing's triad** (late, dangerous): rising BP with wide pulse pressure, **bradycardia**, irregular breathing.

Normal ICP is about 5–15 mmHg; treat above 22 mmHg; aim for CPP 60–70 mmHg (CPP = MAP – ICP).

7 marks Physiotherapy in the acute stage of traumatic brain injury

Asked: S18 (2c), S20 (2e) (asked 2 times)

Principles: protect the injured brain (keep within ICP and CPP limits), prevent secondary complications, match treatment to the level of consciousness, involve the family.

Before each session: GCS, pupils, ICP and CPP, BP, oxygen, sedation, ventilator settings, skull precautions, other fractures.

Area	Physiotherapy
Positioning	Head of bed up about **30°**, head in **midline**; avoid neck flexion and rotation (jugular compression); side lying and supported positions to reduce abnormal postures; turn regularly
Chest care	Positioning, gentle manual techniques, short suctioning with pre-oxygenation; **no head-down drainage** while ICP is raised
Range and tone	Daily passive ROM of all joints; splints or **serial casting** of ankles and wrists when tone rises
Arousal	Sitting over the bed edge and **tilt table** once ICP is controlled; short periods of meaningful input (family voices, music, touch); avoid overstimulation
Skin	Turning schedule, pressure-relieving mattress
Agitation (Rancho IV)	Quiet room, one therapist, short sessions, familiar automatic tasks; low bed and mats instead of restraints
Family	Explain the stages of recovery; teach positioning and range

Stop if ICP rises and stays above the limit, CPP falls, GCS drops or a pupil dilates. After craniectomy, protect the skull defect.

5A(c) Spinal cord disorders: traumatic and non-traumatic

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3 marks Sacral sparing

Asked: S16 (1c)

- **Definition:** preservation of any function in the lowest sacral segments (**S4–S5**) after SCI: light touch or pinprick at the anal margin, **deep anal pressure**, or **voluntary anal contraction**.
- **Why it happens:** sacral fibres lie on the **outer edge** of the long tracts, so they often survive a central lesion.
- **Importance:** it shows some fibres cross the lesion, so the injury is **incomplete** (AIS B–D) with a better prognosis for recovery and walking. No sacral sparing = complete (AIS A).

3 marks Complete vs incomplete paraplegia

Asked: W25 (1c)

Feature	Complete	Incomplete
S4–S5 function	None (no sacral sparing)	**Sacral sparing** present
AIS grade	A	B, C or D
Below the level	No motor or sensory function (except a zone of partial preservation)	Some sensation and/or motor function
Patterns	Symmetrical loss	Cord syndromes: central, anterior, Brown-Séquard, posterior
Outlook	Most stay AIS A; walking with KAFOs for exercise only	Better; many AIS D walk in the community

3 marks Spinal shock and its features

Asked: S17 (1c), W20 (1d) (asked 2 times)

Spinal shock is the temporary loss or depression of all spinal reflex activity below the level of an acute cord injury.

- **Flaccid paralysis** and **areflexia** below the lesion, including the **bulbocavernosus reflex**; loss of sensation.
- **Atonic bladder** with retention; paralytic ileus; loss of sweating and temperature control below the lesion.
- **Phases** (Ditunno, 2004): areflexia (day 0–1); first reflexes return, bulbocavernosus first (days 1–3); early hyperreflexia (1–4 weeks); late hyperreflexia and spasticity (1–12 months).

Not neurogenic shock: that is low BP and bradycardia from loss of sympathetic tone in lesions at or above T6.

3 marks Autonomic dysreflexia (features)

Asked: S17 (1f), W19 (1e), S21 (1d), W21 (1d), S22 (1c), S24 (1d) (asked 6 times)

Definition: a sudden rise in systolic BP of **more than 20 mmHg** above the usual level in SCI **at or above T6**; an emergency (stroke, seizure, death).

- **Cause:** a harmful stimulus below the lesion, most often a **full bladder** (blocked catheter); also bowel impaction, pressure sore, ingrown toenail, tight clothes.
- **Features:** sudden **pounding headache**; high BP; **slow pulse**; flushing and sweating above the lesion; goose bumps; blurred vision; blocked nose; anxiety.
- **First aid:** sit upright, lower the legs, loosen clothes, check BP every 1–2 minutes, check the bladder first, then the bowel; call the doctor.

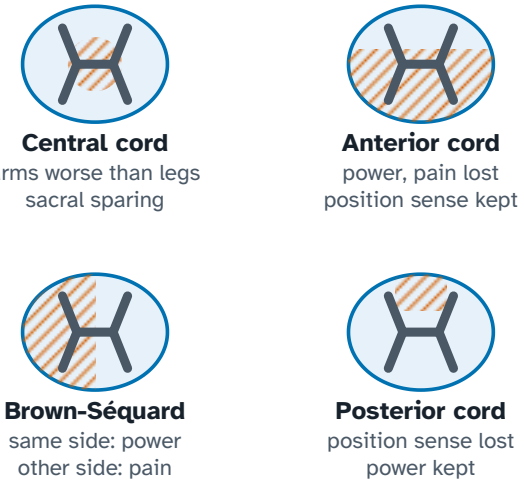
3 marks Brown-Séquard syndrome: features

Asked: W17 (1e), S23 (1c) (asked 2 times)

Hemisection of the cord (stab or bullet wound, tumour, disc).

- **Same side** below the lesion: **weakness with UMN signs** (corticospinal tract); loss of **position sense and vibration** (dorsal columns).
- **Opposite side:** loss of **pain and temperature** from 1–2 segments below (spinothalamic tract crosses).
- At the level, same side: a band of total sensory loss and LMN signs.
- **Prognosis:** the best of the cord syndromes; most walk again.

Incomplete cord syndromes: which part of the cord is damaged



Each oval is a cross-section of the cord, back at the top and front at the bottom; the H shape is the grey matter. Hatching shows the damaged area.

3 marks Cauda equina injury: features

Asked: W24 (1c)

Fracture or disc **below L1–L2** vertebrae damages the nerve roots (not the cord), so the picture is **LMN**:

- **Flaccid**, often **asymmetrical** leg weakness with wasting; lost knee and ankle reflexes.
- **Saddle anaesthesia**; radicular (root) pain.
- **Areflexic (atonic) bladder** and bowel; sexual dysfunction.
- Better outlook than cord lesions: roots are peripheral nerves and can regenerate.

3 marks Paraplegia in flexion and in extension

Asked: W22 (1a)

Feature	In extension	In flexion
Tracts	Mainly **pyramidal**; extrapyramidal (vestibulospinal) still work	Pyramidal **and** extrapyramidal damaged: complete or near-complete
Posture and tone	Legs **extended and adducted**; extensor tone; clonus	Hips and knees **drawn up**; flexor spasms; mass reflex on touch
Outlook	Better (incomplete)	Poorer (severe lesion)

7 marks Autonomic dysreflexia: causes, symptoms and management

Asked: W18 (2d), W24 (2b) (asked 2 times)

Definition: a sudden rise in systolic BP of more than 20 mmHg above the person's usual level (15 mmHg in children) in SCI **at or above T6**. A medical emergency.

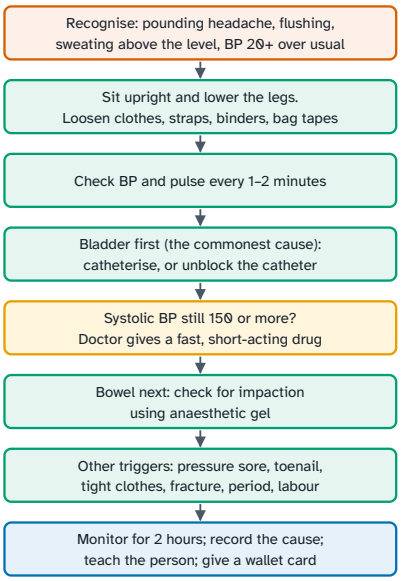
Mechanism

1. A noxious stimulus below the lesion sends impulses up the cord.
2. A massive reflex sympathetic discharge narrows the splanchnic vessels (T6–L2, below the lesion): BP rises sharply.
3. The brain slows the heart through the vagus and dilates vessels above the lesion (flushing, headache).
4. Calming signals cannot pass down the cord, so BP stays high until the trigger is removed.

Causes: bladder distension (commonest: kinked or blocked catheter), bowel impaction, pressure injury, ingrown toenail, tight clothes or straps, fractures, menstruation, labour, sexual activity, procedures.

Symptoms and signs: pounding headache; high BP; bradycardia; flushing and sweating above the lesion; goose bumps; blurred vision; blocked nose; anxiety.

Autonomic dysreflexia: what to do, in order



Order from the 2020 Consortium for Spinal Cord Medicine guideline. A physiotherapist starts the first four steps at once and calls the doctor.

▶ **Text version of the flowchart**

Management (in order)

1. Sit upright, lower the legs; loosen clothes, straps and binders.
2. Check BP and pulse every 1–2 minutes.
3. Check the **bladder** first: catheterise or unblock the catheter.
4. If systolic BP stays ≥150 mmHg, the doctor gives a fast, short-acting drug (nitrate paste, nifedipine).
5. Check the **bowel** with anaesthetic gel; look for other triggers.
6. Monitor for 2 hours; record the cause; educate; give a wallet card.

Prevention in physiotherapy: ask about the bladder before each session; no tight straps on tilt table or standing frame; know the usual BP (90–110 in tetraplegia); stop exercise or stimulation at once if headache starts.

7 marks Ambulation training after an L3 vertebral fracture

Asked: S18 (2e)

An L3 fracture lies **below the end of the cord** (L1–L2), so it damages the **cauda equina**: LMN, often incomplete and

uneven leg weakness, sensory loss and an areflexic bladder. Community walking is expected.

Before walking

- Follow brace and lifting precautions; check skin and insensate feet.
- **Strengthen** hip extensors, abductors and quadriceps (if partially working), and the upper limbs (shoulder depressors, latissimus, triceps) for crutches.
- Keep range: hip flexors, hamstrings, calves.
- Sitting and standing balance; tilt table or frame if needed.

Orthoses

- **AFOs** for weak ankles (foot drop); **KAFOs** only if quadriceps are weak.

Gait training

1. Parallel bars: sit to stand, standing balance, weight shift.
2. Walker or **elbow crutches**: **four-point** gait, then **two-point** gait, then canes.
3. Stairs, kerbs, slopes, rough ground; getting up from the floor.
4. Endurance: 6-minute walk; wheelchair for long distances if needed.

Protect insensate feet (footwear, daily checks); teach safe lifting.

5A(d) Peripheral nerve injuries and the brachial plexus

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3 marks Classification of nerve injuries (Seddon and Sunderland)

Asked: S21 (1c), W21 (1a), S22 (1b), S23 (1a), S25 (1a) (asked 5 times)

Sunderland (Seddon)	Damage	Recovery
I (neurapraxia)	Local myelin damage; conduction block; axon intact; no Wallerian degeneration	Full, days to 3 months
II (axonotmesis)	Axon cut; endoneurium intact; Wallerian degeneration	Full, about 1 mm a day
III	Axon and endoneurium cut; perineurium intact	Incomplete (misdirection)
IV	Only epineurium intact (neuroma in continuity)	Poor without surgery
V (neurotmesis)	Whole nerve divided	None without repair

Seddon has three types; Sunderland splits axonotmesis into degrees II–IV by how much connective tissue is cut.

7 marks Sunderland's classification; physiotherapy for foot drop (4 + 3)

Asked: S16 (2a)

Sunderland's classification (4)

- **Degree I** (neurapraxia): conduction block, axon intact; full recovery in days to 3 months.
- **Degree II** (axonotmesis): axon cut, endoneurium intact; Wallerian degeneration; full recovery at about 1 mm a day.
- **Degree III:** axon and endoneurium cut, perineurium intact; incomplete recovery (misdirection).
- **Degree IV:** only the epineurium intact (neuroma in continuity); needs excision and repair.
- **Degree V** (neurotmesis): nerve divided; no recovery without surgery.

Physiotherapy for foot drop (3)

- **AFO** (posterior leaf spring) to lift the foot in swing and prevent toe catching.
- Daily **calf stretching** to prevent equinus; passive range of the ankle.
- **Re-education** of dorsiflexors and evertors as recovery comes: biofeedback, electrical stimulation, gravity-eliminated then against gravity.
- **Gait training** to replace the high-stepping pattern; protect numb skin on the top of the foot.
- Tendon transfer (tibialis posterior) if no recovery.

7 marks Radial nerve injury (palsy): features and physiotherapy

Asked: S18 (2b), W18 (2e) (asked 2 times)

Causes: fracture of the shaft of the humerus (spiral groove); "Saturday night palsy"; crutch pressure in the axilla.

Features (at the spiral groove)

- **Wrist drop**, finger drop and thumb drop; triceps usually spared.
- Weak grip because the wrist cannot be held extended.
- Sensory loss over a small area on the back of the first web space.
- Posterior interosseous lesion: finger and thumb drop only; wrist extends with radial deviation; no sensory loss.

Physiotherapy

- **Splints:** static **cock-up splint** at night; **dynamic cock-up splint** by day for grasp and release.
- Daily passive ROM of wrist and fingers; prevent shortening of the flexors and overstretching of the extensors.
- **Re-education** as recovery comes (brachioradialis and wrist extensors first, then finger and thumb extensors): biofeedback or electrical stimulation, gravity-eliminated, then active and resisted.
- Grip activities with the wrist supported; task and work practice.
- Explain regrowth time (about 2.5 cm a month); tendon transfer if no recovery.

7 marks Claw hand and its physiotherapy

Asked: W19 (2b)

Claw hand: hyperextension of the **MCP** joints with flexion of the **IP** joints, because the **lumbricals and interossei** are paralysed: the long extensors pull the knuckles back and the long flexors bend the fingers, unopposed.

- **Ulnar lesion:** ring and little fingers clawed; hypothenar and interossei wasting; Froment's sign; card test.
- **Ulnar paradox:** lesion at the elbow claws less, because the long flexors to ring and little fingers are also weak.
- **Median + ulnar:** all four fingers clawed with an ape-thumb.
- Other causes: Klumpke's palsy, leprosy, Volkmann's contracture.

Physiotherapy

- **Knuckle-bender (anti-claw) splint**: blocks MCP hyperextension so the long extensors straighten the IP joints.
- Passive stretching of the MCP joints into flexion; keep the thumb web open.
- **Intrinsic re-education**: finger spreading, "table-top" position, with biofeedback or electrical stimulation.
- Protect the numb little finger; sensory re-education.
- Grip and pinch practice; tendon transfer if no recovery.

7 marks Physiotherapy of carpal tunnel syndrome

Asked: S22 (2c)

CTS: compression of the median nerve under the flexor retinaculum: tingling and numbness of thumb, index, middle and half the ring finger, worse at night; later thenar wasting. Tests: **Phalen's**, **Tinel's** at the wrist, carpal compression.

- **Splint:** neutral wrist splint at night for short-term relief; day wear if night wear alone fails.
- **Education:** avoid sustained wrist bending and forceful gripping; keyboard and mouse habits; breaks.
- **Manual therapy** to the neck and upper limb; superficial heat, interferential or diathermy; splint plus stretching (weak evidence, mild to moderate cases).
- Ergonomic changes to tools and work posture.
- **Do not use** low-level laser, iontophoresis, magnets or thermal ultrasound (APTA 2019); nerve gliding has conflicting evidence.
- **Refer** for injection or surgery if thenar wasting, constant numbness or no response.
- After release: wound care, tendon gliding, grip strengthening.

7 marks Physiotherapy in common peroneal nerve injury

Asked: W25 (2d)

Causes: fibular neck fracture, tight plaster, prolonged squatting or crossed legs, knee dislocation.

Features: foot drop (weak dorsiflexion, eversion, toe extension); high-stepping gait with foot slap; sensory loss over the outer leg and back of the foot.

Physiotherapy

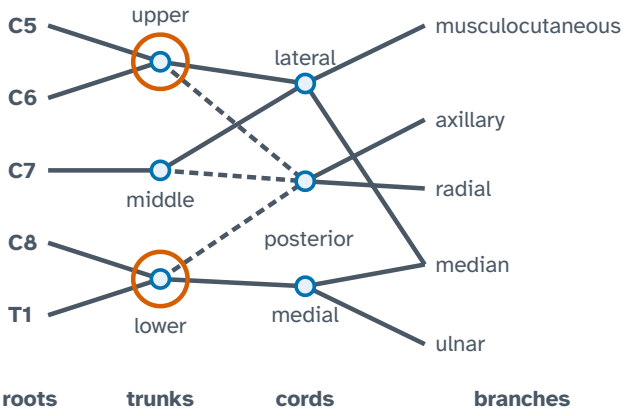
1. **Protect:** posterior leaf spring **AFO**; remove the cause (loosen plaster, avoid crossed legs and squatting).
2. **Range:** daily calf stretching to prevent equinus.
3. **Re-education:** dorsiflexors and evertors with biofeedback or electrical stimulation as Tinel's sign advances; functional electrical stimulation during gait if available.
4. **Strengthening:** gravity-eliminated, then against gravity, then resisted.
5. **Gait training:** heel strike, normal swing; uneven ground.
6. **Skin care** of the numb top of the foot; footwear.
7. Tendon transfer if no recovery.

7 marks Erb's palsy: features and physiotherapy (motor re-education)

Asked: S17 (2c), S21 (2e), W21 (2e), W24 (2a) (asked 4 times)

Erb's palsy: upper plexus (**C5–C6**, upper trunk) injury when the head and shoulder are forced apart (two-wheeler fall; in babies, shoulder dystocia).

The brachial plexus (simplified)



Dashed lines are the posterior divisions. Circles mark the upper trunk (Erb's palsy, C5–C6) and the lower trunk (Klumpke's palsy, C8–T1).

Features

- Lost shoulder abduction (deltoid, supraspinatus), external rotation (infraspinatus), elbow flexion (biceps, brachialis, brachioradialis), supination.
- **"Waiter's tip"** posture: arm adducted and internally rotated, elbow extended, forearm pronated.
- Sensory loss over the outer arm and forearm; biceps reflex lost; risk of shoulder subluxation.

Physiotherapy

- **Support** the arm (sling or shoulder support); prevent subluxation and overstretching.
- **Range:** full shoulder range daily, especially **external rotation and abduction**, and supination.
- **Motor re-education** of deltoid, spinati, biceps and supinator: gravity-eliminated positions first (lying, powder board, sling suspension), with **EMG biofeedback** or electrical stimulation; then against gravity and resisted.
- Functional reaching and task practice; sensory protection.
- Refer to a plexus surgeon if no recovery by about 3–6 months; after nerve transfer, fire the donor muscle to activate the target, then practise alone.

7 marks Klumpke's paralysis: features and physiotherapy

Asked: S25 (2a)

Klumpke's palsy: lower plexus (**C8–T1**) injury when the arm is pulled forcibly overhead (grabbing a branch while falling; breech delivery).

Features

- Paralysis of **hand intrinsics**, finger and wrist flexors: **claw hand**.
- Sensory loss on the inner forearm and hand.

- **Horner's syndrome** (ptosis, miosis, anhidrosis) if the T1 root is avulsed: poor prognosis.

Physiotherapy
- **Anti-claw (knuckle-bender) splint**; keep the thumb web open.
- Passive range of fingers and wrist; MCP flexion stretch.
- **Re-education** of intrinsics and long flexors with biofeedback or electrical stimulation; grip and pinch practice.
- **Sensory** protection and re-education of the inner hand.
- Task practice for work; refer for surgery if no recovery.

5A(d) Peripheral neuropathies and nerve root lesions

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7 marks Diabetic neuropathy: classification; physiotherapy and care of the diabetic foot

Asked: W17 (2c), W19 (2f), W22 (2e), S23 (2a) (asked 4 times)

Classification (ADA, 2017)
- **Diffuse: distal symmetric polyneuropathy** (commonest, "glove and stocking"); **autonomic** neuropathy (heart, BP, gut, bladder, sweating).
- **Mononeuropathy:** single cranial or limb nerve (third nerve, median, peroneal); mononeuritis multiplex.
- **Radiculopathy:** lumbosacral radiculoplexus neuropathy (diabetic amyotrophy); thoracic radiculopathy.

Diabetic foot = neuropathy + peripheral artery disease + deformity → ulcer, infection, amputation. Loss of protective sensation, weak intrinsics (clawed toes, high pressure under metatarsal heads), dry cracked skin.

Assessment
10 g monofilament, 128 Hz tuning fork, foot pulses and ABI, deformity, skin and nails, balance; IWGDF risk category (0 very low to 3 high: screen yearly to every 1–3 months).

Physiotherapy and foot care
- **Daily inspection** with a mirror; wash, dry between toes, moisturise (not between toes); nail and callus care.
- **Protective footwear** that fits; **no barefoot walking**; check shoes before wearing.
- **Foot-ankle exercises**: toe flexion and extension, ankle range and strength.
- **Balance training** with vision, then varied surfaces; good home lighting.
- **Graded walking** in proper footwear (extra 1000 steps a day); cycling or swimming if an ulcer is present.
- **Ulcer:** offloading (total contact cast, non-removable walker), crutches or wheelchair.
- Burning pain: graded activity, TENS trial.

Precautions: no heat or ice over numb feet; check feet after exercise; check sugar before exercise; a hot, swollen painless foot may be Charcot foot: refer the same day.

7 marks Leprous neuritis; features of lepromatous leprosy

Asked: W22 (2b)

Leprosy: chronic infection with *Mycobacterium leprae* affecting skin and peripheral nerves. Spectrum (Ridley–Jopling): tuberculoid → borderline → lepromatous.

Lepromatous leprosy
- Many **symmetrical** skin lesions and nodules; thickened facial skin (**leonine face**); loss of eyebrows; collapsed nose.
- **Glove-and-stocking** anaesthesia; symmetrical nerve thickening; eye damage.
- Skin smears strongly positive (multibacillary).

Leprous neuritis
- A **painful, tender, swollen nerve** with new loss of sensation or power, often during a **reaction** (type 1 reversal; type 2 ENL). "Silent neuritis": loss without pain.
- Nerves: ulnar (elbow), common peroneal (fibular neck), posterior tibial, median, facial, great auricular.
- Deformities: claw hand, foot drop, clawed toes, plantar ulcers, lagophthalmos.

Management
- WHO multidrug therapy (rifampicin, dapsone, clofazimine) 6 or 12 months; oral steroids for neuritis.
- **Physiotherapy: rest** the limb in a splint during acute neuritis (no exercise of the inflamed nerve); then gentle range and graded exercise; **monitor nerve function** (voluntary muscle and monofilament testing) every visit; knuckle-bender splint for claw hand; AFO and microcellular rubber footwear; self-care "**soak, scrape, oil**"; padded tools and gloves; rehabilitation after tendon transfer; counselling against stigma.

7 marks Alcoholic neuropathy and its physiotherapy

Asked: S25 (2f)

Definition: a length-dependent, mainly axonal **sensorimotor polyneuropathy** from long-term heavy drinking, worsened by poor nutrition (especially **thiamine** deficiency).

Features
- Burning, painful feet; numbness in a glove-and-stocking pattern; lost ankle reflexes.
- **Sensory ataxia** (worse in the dark; Romberg positive).
- Distal weakness and wasting; foot drop in severe cases; autonomic problems.

Medical: stop alcohol; thiamine and vitamins; good diet; medicines for nerve pain. Recovery is slow and often partial.

Physiotherapy
- **Balance and ataxia:** balance training with visual feedback; **Frenkel's exercises**; a stick; good lighting.
- **Weakness:** progressive strengthening of ankle and hand muscles; **AFO** for foot drop.
- **Pain:** graded activity, TENS trial, education.
- **Foot protection:** daily inspection, proper footwear (as for the diabetic foot).
- General fitness; support to stay off alcohol.

Precautions: withdrawal, low sugar, liver disease; protect numb skin from heat.

7 marks Physiotherapy for neuropathic pain

Asked: W21 (2a)

Definition (IASP): pain caused by a lesion or disease of the **somatosensory nervous system**.

- **Features:** burning, shooting or electric pain; pins and needles; **allodynia** (pain from light touch); **hyperalgesia**; numbness in the same area.
- **Causes:** diabetic neuropathy, post-herpetic neuralgia, trigeminal neuralgia, nerve injury, radiculopathy, SCI, stroke, MS, phantom limb.
- **Medicines** (doctor): gabapentin or pregabalin, duloxetine, amitriptyline.

Physiotherapy

- **Pain education** in plain words: hurt does not always mean harm; nerves become over-sensitive.
- **Graded activity** and regular exercise with **pacing** (avoid boom-and-bust).
- **Desensitisation** for allodynia: graded textures and pressure.
- **Sensory discrimination** training; **graded motor imagery** and **mirror therapy** (CRPS, phantom pain).
- **TENS** for short-term relief; relaxation, sleep advice.

7 marks Physiotherapy of reflex sympathetic dystrophy (CRPS)

Asked: S21 (2a)

CRPS: a painful limb condition, usually after injury, surgery or immobilisation (often a wrist fracture), with pain **out of proportion** to the injury and sensory, vasomotor, sudomotor, oedema, motor and trophic changes. **Type I** (RSD): no nerve injury; **type II** (causalgia): with nerve injury.

Budapest criteria: continuing disproportionate pain; symptoms in 3 of 4 and signs in 2 of 4 groups: sensory (allodynia), vasomotor (colour, temperature), sudomotor/oedema, motor/trophic (stiffness, weakness, hair and nail change); no other diagnosis.

Physiotherapy

- Early, gentle **movement and functional use**; avoid prolonged immobilisation.
- **Graded motor imagery:** left/right recognition → imagined movement → **mirror therapy**.
- **Desensitisation** and tactile discrimination training.
- **Oedema:** elevation, active pumping, retrograde massage or compression if tolerated.
- Graded exercise and return to activity and work; pacing.
- Team: medicines, psychology, occupational therapy.

Do not force painful passive stretching; never splint for long without movement.

5A(e) Vestibular disorders

Full notes p. 116

7 marks Signs and symptoms of vestibular lesions; adaptation exercises

Asked: S17 (2e)

Signs and symptoms

- **Vertigo:** false sense of spinning; dizziness.
- **Nystagmus:** peripheral lesions give horizontal-torsional nystagmus beating away from the damaged side, reduced by fixing the gaze.
- **Imbalance:** unsteady standing and walking, veering or falling to the damaged side.
- **Oscillopsia:** the world jumps during head movement (VOR failure).
- Nausea, vomiting, sweating; hearing loss, tinnitus (Ménière's, labyrinthitis); anxiety and avoidance of head movement.

Central signs (refer): direction-changing or vertical nystagmus, normal head impulse test, skew deviation, double vision, slurred speech, weakness.

Adaptation (gaze stabilisation) exercises

- **Principle:** image slip on the retina during head movement drives the cerebellum to **recalibrate the VOR** (increase gain).
- **VOR ×1:** eyes fixed on a stationary target (a letter at arm's length) while moving the head side to side, then up and down, for 1–2 minutes.
- **VOR ×2:** target and head move in opposite directions while the eyes stay on the target.
- **Progress:** speed, duration, busy background, sitting → standing → walking.
- **Dose** (Hall 2022): acute one-sided loss 3 times a day, at least 12 minutes daily; chronic 3–5 times a day, at least 20 minutes, for 4–6 weeks. Mild symptoms during exercise are expected.

7 marks Tests for vestibular disorders (explain two)

Asked: W24 (2c)

Enumerate: head impulse test; dynamic visual acuity; Dix-Hallpike; supine roll test; head-shaking test; Romberg and sharpened Romberg; Fukuda stepping test; mCTSIB; Dizziness Handicap Inventory; Motion Sensitivity Quotient; Dynamic Gait Index; laboratory tests (caloric, video head impulse, VEMP).

1. Head impulse test (VOR)

- The person fixes on the examiner's nose; the examiner turns the head quickly and unpredictably 10–20° to each side.
- **Positive:** a **catch-up saccade** back to the target shows a weak VOR on the side the head was turned towards.

2. Dix-Hallpike test (posterior canal BPPV)

- Sitting, head turned 45° to one side; lie back quickly with the neck extended about 20° below horizontal.
- **Positive:** vertigo with **upbeating, torsional nystagmus** after a short delay, lasting under a minute: BPPV on the lower ear.

Other good choices: Fukuda (turning more than 45° in 50 steps towards the lesion); dynamic visual acuity (a drop of more than two lines with head movement).

7 marks BPPV: diagnostic test for the posterior canal and physiotherapy

Asked: S20 (2f), W21 (2c), S26 (2d) (asked 3 times)

BPPV: otoconia come loose from the utricle into a semicircular canal (posterior most often); head movement shifts them, causing **brief vertigo** (seconds) on lying down, turning in bed, looking up or bending; no hearing loss.

Diagnosis: Dix-Hallpike test

From sitting with the head turned 45° to one side, bring the person quickly to lying with the neck extended 20° (affected ear down). Positive: vertigo with **torsional, upbeating nystagmus** after a short latency, lasting under a minute. If negative, test the other side; if horizontal nystagmus, do the supine roll test (horizontal canal).

Epley manoeuvre (right posterior canal)

1. Sit, head turned 45° to the right.
2. Lie back quickly with the neck extended 20° (right Dix-Hallpike); wait until vertigo stops (≥30 s).
3. Turn the head 90° to the left; wait 30 s.
4. Roll onto the left side, head turned a further 90° (nose down); wait 30 s.
5. Sit up with the chin slightly down; repeat Dix-Hallpike and repeat if positive.

Alternatives: Semont (liberatory) manoeuvre; barbecue roll or Gufoni for horizontal canal; Brandt–Daroff exercises at home for residual symptoms.

Advice (BPPV guideline, 2017): no post-manoeuvre postural restrictions; no routine suppressant drugs; reassess within a month; teach recurrence and fall risk.

> **Draw:** The Dix-Hallpike position: head turned 45°, neck extended 20° over the couch end.

7 marks Vestibular rehabilitation: principles and manoeuvres

Asked: S22 (2e), S24 (2b) (asked 2 times)

Principles

- Offer exercise-based rehabilitation to adults with one- or two-sided vestibular hypofunction; start **early** (within 2 weeks of acute loss).
- Recovery comes from **adaptation** (recalibrating the VOR), **habituation** (reducing response to provoking movements) and **substitution** (other senses and strategies).
- Exercises should provoke **mild** symptoms; stop vestibular suppressants early (they slow compensation).
- Supervised sessions plus a daily home programme; progress step by step; add balance and gait training.

Exercises

- **Gaze stabilisation:** VOR ×1 and ×2, progressed in speed, background and position.
- **Habituation:** repeat the movements that provoke mild dizziness (Motion Sensitivity Quotient), twice a day.
- **Balance and gait:** narrowing base, foam, eyes closed; walking with head turns, pivots, dual tasks.
- **Dose:** chronic one-sided loss: gaze 3–5 times a day (≥20 min) and balance ≥20 min daily, 4–6 weeks.

Two manoeuvres (BPPV)

1. **Epley:** Dix-Hallpike position → head turned 90° to the other side → roll onto that side, nose down → sit up; about 30 s in each position.
2. **Semont:** head turned away from the affected ear; move quickly onto the affected side (1 min), then swing through to the other side, nose down (1 min); sit up slowly.

Do not use eye exercises without head movement (classic Cawthorne–Cooksey start) to improve gaze stability.

7 marks Substitution and habituation exercises

Asked: W25 (2e)

Habituation

- **Principle:** repeated, controlled exposure to movements or visual scenes that provoke **mild** dizziness reduces the response over time (central nervous system learns to ignore it).
- **Method:** find provoking movements with the **Motion Sensitivity Quotient** (sit to supine, rolling, bending, head turns); choose 3–4 that give mild to moderate symptoms; repeat each a few times, **twice a day**, letting symptoms settle between repetitions.
- Examples: **Brandt–Daroff** exercises; head turns while walking; optokinetic (busy visual) exposure.
- **Used for:** motion-provoked dizziness, visual motion sensitivity, persistent postural-perceptual dizziness.

Substitution

- **Principle:** replace lost vestibular function with **vision**, **somatosensation** and planned strategies, when the loss is **bilateral or complete**.
- **Methods:** balance training that uses vision and feet (firm surfaces, then progressively harder); **planned eye–head movements** (look at the target first, then turn the head); central pre-programming of eye movements; use of the cervico-ocular reflex.
- **Strategies:** a stick, night lights, avoid dark and uneven ground, turn slowly; fall prevention at home.

Third type, adaptation (VOR ×1, ×2), is for one-sided loss; most programmes combine all three with gait training.

5A(f) Facial (VII) nerve palsy and other cranial nerves

Full notes p. 124

3 marks Clinical assessment of the facial nerve

Asked: S16 (1f)

Test	Muscle or function
Raise eyebrows, wrinkle forehead	Frontalis (**spared in UMN** palsy)
Close eyes tightly against resistance	Orbicularis oculi; watch Bell's phenomenon
Show teeth, smile	Zygomaticus, levators of the lip
Puff cheeks, whistle	Buccinator, orbicularis oris
Tense the neck	Platysma
Taste, tears, hearing	Front two-thirds of tongue; Schirmer's test; stapedius (sound sensitivity)

Grade with **House-Brackmann** (I normal to VI total paralysis) or Sunnybrook.

3 marks Signs and symptoms of Bell's palsy

Asked: W24 (1d)

Acute, one-sided **LMN** facial palsy of unknown cause, developing over less than 72 hours.

- Weakness of the **whole half** of the face: cannot raise the eyebrow or wrinkle the forehead; eye does not close (**Bell's phenomenon** visible); flat nasolabial fold; mouth droops and pulls to the normal side.
- Food collects in the cheek; drinks dribble; unclear lip sounds (p, b, m).
- Pain behind the ear; watering or dry eye; loss of taste (front two-thirds of tongue); **hyperacusis**.

3 marks Bell's phenomenon

Asked: W18 (1e)

- **Definition:** when the patient tries to close the eyes, the eyeball on the weak side **rolls upwards and outwards**.
- It is a **normal protective reflex** that becomes **visible** in LMN facial palsy because the eyelid (orbicularis oculi) cannot close over it.
- **Importance:** shows incomplete eye closure, so the cornea is at risk: eye drops, ointment, taping at night, glasses. A poor Bell's phenomenon means greater corneal risk.

3 marks Facial synkinesis

Asked: W18 (1d)

- **Definition:** involuntary movement of one part of the face during voluntary movement of another (the eye narrows on smiling; the mouth corner lifts on blinking).
- **Cause: misdirected regrowth** of facial nerve axons after severe (axonal) palsy; appears months later. "Crocodile tears" (tears while eating) are a similar misdirection.
- **Management:** slow, small, isolated, symmetrical movements with a **mirror or EMG biofeedback** (neuromuscular retraining); relax the unwanted muscle; massage and stretching; avoid forceful mass movements and gum chewing; botulinum toxin by the doctor.

3 marks Clinical assessment of the trigeminal nerve

Asked: W18 (1c)

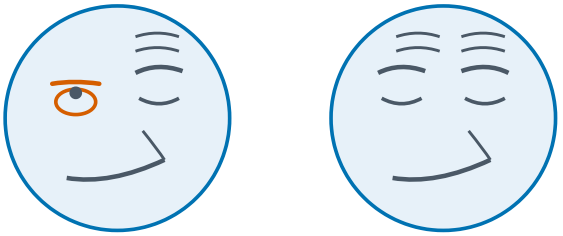
- **Sensory:** light touch and pinprick over the forehead (**V1** ophthalmic), cheek (**V2** maxillary) and jaw (**V3** mandibular), comparing sides.
- **Motor:** clench the teeth and feel masseter and temporalis; open the mouth against resistance: jaw deviates to the **weak side** (pterygoids).
- **Reflexes: corneal reflex** (V1 afferent, facial efferent); **jaw jerk** (brisk in UMN lesions above the pons).

7 marks Bell's (facial) palsy: features and physiotherapy, with education and advice

Asked: W16 (2a), W17 (2a), S24 (2f) (asked 3 times)

Bell's palsy: acute, one-sided LMN facial palsy, probably viral inflammation swelling the nerve in its bony canal. **Features:** whole half of the face weak (forehead included), eye does not close with Bell's phenomenon, mouth droop, dribbling, loss of taste, hyperacusis.

Right facial weakness: LMN vs UMN, on trying to close the eyes and smile



LMN (Bell's palsy)
whole half of the face

UMN (stroke)
lower face only

The faces are drawn as you see the patient, so the patient's right side is on your left. The forehead is spared in UMN palsy because the upper face receives nerve supply from both sides of the brain.

Medical: oral steroids **within 72 hours** (recovery 83% vs 64% at 3 months); eye protection; refer if no recovery by 3 months.

Physiotherapy by stage

- **Acute (paralysed):** reassurance; **eye care**; gentle upward massage towards the eye and ear; moist warmth with the eye protected; cheek support taping.
- **Recovering:** active-assisted then active **mirror exercises** for each group: raise eyebrows, close eyes gently, smile, purse lips, puff cheeks, flare nostrils. Slow, small, **symmetrical**; hold the strong side back.
- **Late: neuromuscular retraining** with mirror or EMG biofeedback for synkinesis; stretch and massage tight muscles.

Education and advice

- Most recover in weeks to months.
- Eye: drops by day, ointment and taping at night, glasses outdoors; report pain or redness.
- Chew on the strong side; clear food from the cheek; sip through a straw.
- Exercise gently several times a day; avoid forceful grimacing and gum chewing.

Electrical stimulation shows no benefit over placebo (Cochrane 2011) and may encourage mass movement.

7 marks Trigeminal neuralgia: features and physiotherapy

Asked: S20 (2a), W22 (2f) (asked 2 times)

Features

- Sudden, severe, **electric shock-like** pain on one side of the face, usually V2 or V3, lasting seconds to 2 minutes, in repeated attacks.
- Triggered by light touch, talking, chewing, brushing teeth, shaving or cold wind (**trigger zones**).
- Normal sensation between attacks; weight loss and poor oral hygiene from avoiding triggers.
- **Cause:** usually a blood vessel compressing the root; MS or tumour in young people or with sensory loss.

Medical: carbamazepine (first choice) or oxcarbazepine; microvascular decompression if drugs fail.

Physiotherapy (supportive)

- **Education** on avoiding triggers: scarf in cold wind, lukewarm water, soft toothbrush, chew on the other side.
- **Relaxation** and stress management; posture and jaw relaxation.
- **TENS** may be tried away from trigger zones.
- Encourage nutrition and oral hygiene.
- **Do not** stimulate trigger zones directly.

5A(g) Multiple sclerosis

Full notes p. 133

3 marks Three clinical features (symptoms) of multiple sclerosis

Asked: W16 (1e), S22 (1f) (asked 2 times)

1. **Visual:** optic neuritis (painful eye movement, blurred central vision, faded colours), double vision, nystagmus.
2. **Sensory:** numbness, tingling, band-like feelings; **Lhermitte's sign** (electric shock down the spine on neck flexion).
3. **Motor:** weakness and spasticity of the legs with UMN signs.

Also: **fatigue** (often the most disabling), cerebellar ataxia and intention tremor, bladder urgency, cognitive and mood changes, heat sensitivity (Uhthoff's).

3 marks Uhthoff's phenomenon

Asked: W22 (1e)

- **Definition:** a **temporary worsening** of MS symptoms (often blurred vision or weakness) when **body temperature rises**: exercise, hot bath, hot weather, fever.
- **Mechanism:** heat slows or blocks conduction further in demyelinated fibres; settles on cooling. It is **not a relapse**.
- **Physiotherapy:** pre-cooling (cool drink, cool shower), cool room, fan, cooling vest, breaks; exercise at the coolest time; avoid hot packs. Cool rather than stop exercise.

3 marks Charcot's triad in multiple sclerosis

Asked: W25 (1b)

1. **Nystagmus**
2. **Intention tremor**
3. **Scanning (staccato) speech**

These are **cerebellar and brainstem** signs from plaques in the cerebellum and its connections. Physiotherapy for the ataxia: proximal stability, Frenkel's exercises, slow controlled practice with visual feedback, weighted wrist cuffs for some tasks.

Do not confuse with Charcot's triad of cholangitis (fever, jaundice, right upper quadrant pain).

7 marks Visual, cognitive and behavioural disturbances in multiple sclerosis

Asked: W18 (2a)

Area	Features	Physiotherapy approach
Visual	**Optic neuritis** (painful eye movement, blurred central vision, poor colour vision); double vision (internuclear ophthalmoplegia); **nystagmus**; oscillopsia	Good lighting; high-contrast step edges; uncluttered spaces; refer for eye care; gaze stability exercises if appropriate
Cognitive	Slowed processing; poor memory, attention and planning (executive function)	Short, simple instructions; **written programmes**; practise in quiet surroundings; avoid dual tasks at first
Behavioural and emotional	**Depression**, anxiety, irritability, emotional lability (sudden laughing or crying), euphoria, low motivation	Agree goals together; encourage; exercise improves mood; refer for counselling

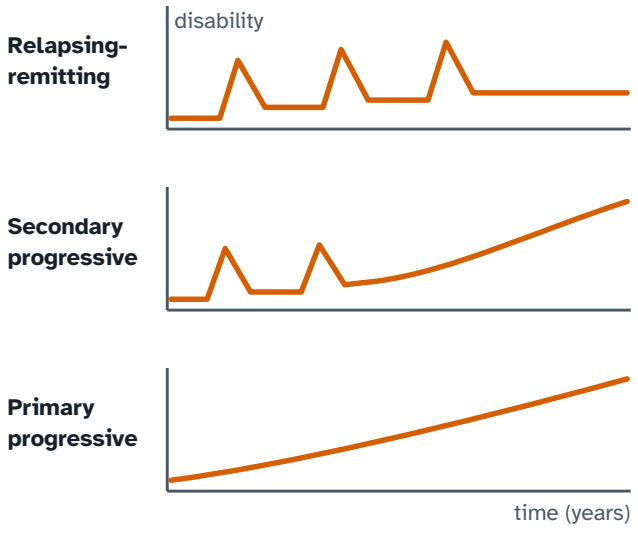
Fatigue and heat (Uhthoff's) make all three worse, so schedule sessions in the cool part of the day and pace activity.

These "invisible" symptoms often limit work and family life more than weakness; they decide how you teach the home programme.

7 marks Stages (courses) of multiple sclerosis; physiotherapy for one stage

Asked: S20 (2c)

Courses of MS: disability over time



About 85% start as relapsing-remitting MS. After years, many move into a secondary progressive course. A minority are progressive from the start.

Course	What happens
Clinically isolated syndrome	A first attack (optic neuritis) that may become MS

Course	What happens
Relapsing-remitting (85% at onset)	Relapses for days to weeks, then full or partial recovery; stable between
Secondary progressive	After a relapsing course, steady increase in disability, with or without relapses
Primary progressive	Steady worsening from the start, often walking difficulty

Physiotherapy in relapsing-remitting MS (EDSS 0–6.5)

- **During a relapse:** rest the affected function, keep range, position well, gentle activity; avoid overheating.
- **After a relapse:** grade back up: transfers, balance, walking, strength.
- **Aerobic:** 2–3 times a week, 10–30 minutes, moderate (40–60% HRmax, RPE 11–13).
- **Resistance:** 2–3 times a week, 1–3 sets of 8–15, 5–10 exercises.
- **Balance and gait** training; AFO for foot drop; stretching for spasticity.
- **Fatigue**: energy conservation; **heat**: pre-cooling. Target 150 minutes a week of activity.

7 marks Multiple sclerosis: features and physiotherapy

Asked: S19 (2c), W20 (2b) (asked 2 times)

MS: chronic immune-mediated demyelination of the CNS, lesions scattered in space and time; average onset 32 years; women twice as often.

Features: optic neuritis, double vision; numbness, Lhermitte's sign; spastic weakness of the legs; ataxia, intention tremor; **fatigue**; bladder urgency; cognitive and mood changes; heat sensitivity (Uhthoff's).

Physiotherapy

- **Principle:** exercise is safe and helpful at every level (Kalb 2020); grade to EDSS and fatigue; keep cool.
- **Aerobic** (EDSS 0–6.5): 2–3 times a week, 10–30 min, moderate intensity; **resistance** 2–3 times a week, 1–3 sets of 8–15.
- **Balance and gait:** static and dynamic balance, stepping, turning; AFO, walking aids.
- **Spasticity:** daily stretching, standing, positioning; check triggers (bladder, skin, infection).
- **Ataxia and tremor:** proximal stability, Frenkel's, weighted cuffs.
- **Fatigue:** plan, prioritise, pace; rest breaks. **Heat:** pre-cooling, cooling vest.
- **Severe (EDSS 7–9):** short daily sessions, arm exercise, standing frame, transfers, seating, carer training.
- **Bladder:** timed voiding, pelvic floor training. Work adjustments.

A new symptom lasting over 24 hours without fever or heat may be a relapse: refer.

5A(g) Guillain–Barré syndrome

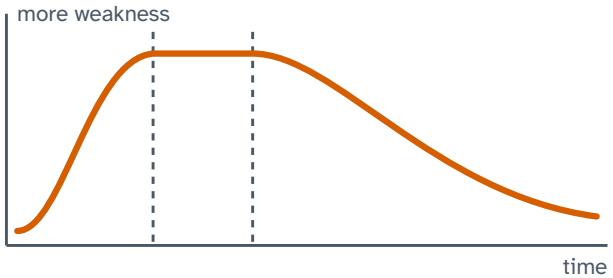
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7 marks Physiotherapy in acute-stage Guillain-Barré syndrome

Asked: W21 (2d)

GBS: acute immune-mediated polyradiculoneuropathy: rapidly **ascending, symmetrical, flaccid** weakness with **areflexia**, worst within 4 weeks; breathing, cranial and autonomic nerves may be involved (up to 22% need a ventilator).

The course of GBS and the physiotherapy focus in each phase



Progression	Plateau	Recovery
up to 4 weeks	days to weeks	weeks to years
chest care	chest care	strengthening
position, ROM	tilt, sit up	gait, fitness

Most people are at their weakest within 2 weeks. Recovery is fastest in the first year but can continue for more than 5 years.

Aims in the ICU (progression and plateau)

Protect breathing, prevent the complications of immobility, monitor the autonomic system, relieve pain, reassure.

Problem	Physiotherapy
Weak breathing muscles	Measure **vital capacity** or single breath count each shift; report VC <20 mL/kg (20/30/40 rule); deep breathing, incentive spirometry, positioning
Secretions	Positioning, manual techniques, suctioning, **assisted cough**; watch for arrhythmia during suctioning
Contracture	Gentle passive ROM once or twice a day; **resting splints** for wrist and hand; AFO or foot support
Skin and pressure palsies	Turn regularly; pad elbows (ulnar) and fibular heads (peroneal)
Autonomic instability	Monitor HR, rhythm, BP during turning and sitting; slow position changes
Pain and anxiety	Gentle handling, positioning, explain before touching
Early mobility	Sitting up and tilt table once breathing and BP are stable

Do not overstretch flaccid joints; support the shoulders; upright for meals if swallowing is weak; protect the eyes if the face is weak.

7 marks Physiotherapy management of Guillain-Barré syndrome

Asked: S19 (2a)

Principles: watch breathing and autonomic system; prevent immobility complications; exercise within the person's strength; **never overwork** recovering muscles; plan return to home and work.

1. Acute (ICU)

- Chest care: positioning, deep breathing, airway clearance, assisted cough; monitor VC or single breath count.
- Passive ROM, resting splints (wrist, ankle), turning, padding of elbows and fibular heads.
- Tilt table and sitting once stable.

2. Early recovery (ward)

- Active-assisted then active exercise, gravity eliminated first.
- Sitting balance, rolling, bridging, sit to stand, standing in parallel bars; short sessions with rests.

3. Rehabilitation

- **Progressive resistance**, few repetitions, rests between sets; stop before fatigue.
- **Gait training**: parallel bars → walker → stick; **AFO** for foot drop.
- **Balance** with visual feedback when position sense is poor.
- **Endurance**: cycling 30 minutes at 70–90% HRmax, 3 times a week for 12 weeks (Garssen 2004); fatigue management (pacing).
- Hand function; return to school or work in stages.

Overwork rule: if weakness or tiredness is worse the next day, the session was too hard: shorten it or lower the load. About 80% walk independently at 6 months.

5A(h) Cerebellar disease and ataxia

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3 marks Rebound test

Asked: W17 (1b)

- **Method:** the patient flexes the elbow against the examiner's resistance; the examiner **suddenly lets go** while guarding the patient's face with the other hand.
- **Normal:** the triceps checks the movement at once.
- **Positive (cerebellar lesion):** the arm flies towards the face or shoulder: impaired check, **Holmes' rebound phenomenon**, on the side of the lesion.
- Variant: tap the outstretched arms downward and watch for excessive swing.

3 marks Assessment of sensory ataxia

Asked: S19 (1f)

- **Romberg's test:** steady with feet together and eyes open; sways or falls with **eyes closed** (positive).
- **Joint position sense** and **vibration** (128 Hz) reduced, starting in the toes.
- Finger-nose and heel-shin tests much **worse with eyes closed**.
- **Pseudoathetosis:** outstretched fingers wander with eyes closed.
- **Gait:** high-stepping, stamping; watches the feet; worse in the dark.

Cause: posterior columns, dorsal roots or peripheral nerves (B12 deficiency, tabes, neuropathy).

7 marks Physiotherapy (treatment) of cerebellar ataxia

Asked: S16 (2e), W16 (2e) (asked 2 times)

Cerebellar ataxia: dysmetria, dysdiadochokinesia, intention tremor, decomposition, rebound, hypotonia; truncal sway; wide-based staggering gait; nystagmus, dysarthria. Assess with SARA, Berg, TUG.

Principles

- **Intensive, repeated** coordination and balance practice: 4 weeks of coordinative training reduced ataxia and improved gait, gains kept with home practice (Ilg 2009).
- Train the person's own goals as whole tasks; increase difficulty gradually; lifelong in progressive disease.

Techniques

- **Proximal stability:** prone on elbows, four-point kneeling, kneeling, half-kneeling, standing; approximation; PNF **rhythmic stabilisation** and stabilising reversals.
- **Balance:** sitting then standing, reducing base, reaching, weight shift, stepping, perturbations.
- **Coordination:** slow, controlled movements to targets, gradually faster; mirror or video feedback; Frenkel's for the sensory component.
- **Gait:** slightly wide base then narrowing; walking on lines, over obstacles, turning; treadmill; rollator or weighted walker.
- **Upper limb:** task practice (writing, eating, buttoning) with elbows stabilised; weighted cutlery and pens, wrist weights.
- **Fitness:** stationary or recumbent cycling.

Guard with a gait belt; avoid fatigue (worsens ataxia); no alcohol or sedatives; home safety.

7 marks Cerebellar ataxia and the rationale of balance training

Asked: W19 (2d)

Cerebellar ataxia: uncoordinated movement (errors of direction, range, timing and force) not due to weakness, from lesions of the cerebellum or its connections. The cerebellum compares intended and actual movement and corrects errors; when it fails, movements overshoot, decompose and tremble near the target.

- **Hemisphere lesion:** same-side limb ataxia (dysmetria, intention tremor, dysdiadochokinesia, rebound).
- **Vermis lesion:** truncal sway (titubation), wide-based staggering gait, cannot tandem walk.
- Nystagmus, scanning dysarthria; Romberg usually negative (unsteady even with eyes open).

Rationale of balance training

- The cerebellum **adapts movement through practice and error**; repeated balance challenges let remaining cerebellar circuits and other areas learn to reduce sway and correct errors (plasticity).
- Strengthens postural muscles and improves **proximal stability**.
- Builds **confidence**, reduces fear of falling and **falls** (balance exercise >3 hours a week prevents most falls, Sherrington 2017).

- Because cerebellar learning is slower, training needs **more repetitions** and longer programmes.

Progression
Sitting → kneeling → standing; wide → narrow base; static → dynamic (reaching, stepping, perturbations); eyes open → closed; firm → soft surface; single → dual task.

5A(i) Parkinson's disease and other extrapyramidal disorders

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3 marks Three functions of the basal ganglia

Asked: W21 (1c)

1. **Select and start** the wanted movement and suppress unwanted ones.
2. **Scale** the size, speed and force of movement.
3. Run well-learned movements **automatically and in sequence** (walking, writing, turning in bed).

Also: motor learning and habits; loops for thinking, motivation and mood. Loss of nigral dopamine (PD) gives slow, small movements and loss of automaticity; other lesions give chorea, athetosis or dystonia.

3 marks Cardinal features of Parkinson's disease

Asked: S25 (1f)

Memory hook **TRAP**:

- **Tremor** at rest, 4–6 per second, "pill-rolling"; starts in one hand; less with movement, absent in sleep.
- **Rigidity:** lead-pipe, or cogwheel when tremor is added; flexors and extensors equally.
- **Akinesia/bradykinesia:** slowness, difficulty starting, movements shrink as they repeat; mask face, soft voice, micrographia.
- **Postural instability:** poor balance reactions, a later feature; falls backwards on the pull test.

MDS (2015): diagnosis needs bradykinesia plus rest tremor or rigidity.

3 marks Rigidity and its types

Asked: W19 (1f), S24 (1b) (asked 2 times)

Rigidity: increased resistance to passive movement, present through the **whole range**, in **flexors and extensors equally**, **independent of speed**; an extrapyramidal sign.

- **Lead-pipe:** smooth, even resistance throughout the range (PD).
- **Cogwheel:** jerky, ratchet-like resistance: tremor superimposed on lead-pipe (PD).
- **Decorticate:** arms flexed, legs extended; lesion above the red nucleus.
- **Decerebrate:** arms and legs extended, arms internally rotated; brainstem lesion (worse prognosis).

3 marks Spasticity vs rigidity

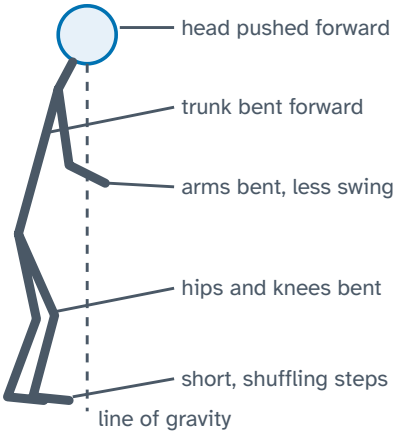
Asked: W17 (1d)

Feature	Spasticity	Rigidity
Lesion	UMN (pyramidal)	Extrapyramidal (basal ganglia)
Speed	**Velocity-dependent**	Same at any speed
Pattern	**Clasp-knife**	**Lead-pipe or cogwheel**
Muscles	Antigravity (arm flexors, leg extensors)	Flexors and extensors equally
Reflexes	Brisk, clonus, Babinski	Normal; no Babinski

7 marks Parkinsonian (festinating) gait: abnormalities and gait training

Asked: S16 (2c), W17 (2e), W16 (2c), S17 (2f) (asked 4 times)

Parkinsonian posture and gait



The dashed line of gravity falls in front of the feet, so the person takes faster and faster short steps to keep up with it (festination).

Gait abnormalities

- **Flexed posture**: head forward, rounded shoulders, hips and knees bent.
- Short, **shuffling** steps, flat foot (no heel strike), poor clearance.
- **Festination:** the line of gravity falls in front of the feet, so steps become shorter and faster as if chasing it; propulsion and retropulsion.
- Reduced arm swing and trunk rotation; **turning en bloc**.
- **Start hesitation and freezing** at doorways, turns, narrow spaces, or with a second task.

Gait training

- **External cues:** floor stripes at target step length; metronome or music; verbal cue "big step". Cues bypass faulty basal ganglia timing.
- **Attention** on long steps and heel strike; no dual tasks early.
- Practise long steps with arm swing; treadmill; Nordic poles; **wide-arc turns**; doorways; stops and starts; backwards and sideways.
- **Festination:** stop, stand tall, restart with a long step; widen the base.
- **Freezing:** stand tall, shift weight, step over a line or mark, count or march on the spot.
- Trunk extension and rotation; stretch hip flexors and hamstrings.

- **Dose:** 20–60 min, 3–5 days a week, 4–12 weeks (APTA 2022); in the "on" period.

Train in the "on" period; monitor with RPE (heart rate response may be blunted).

7 marks Physiotherapy for bradykinesia and rigidity in Parkinson's disease

Asked: S22 (2d)

Bradykinesia: slowness and progressively smaller movements (hypokinesia), difficulty starting. **Rigidity:** lead-pipe or cogwheel resistance in flexors and extensors, causing a stiff flexed trunk.

Bradykinesia

- **Large-amplitude training** (LSVT BIG: 16 one-hour sessions, 4 days a week for 4 weeks): whole-body movements at maximal size.
- **Speed and power** training (40% 1RM fast).
- **External cues** (visual, auditory) and **cognitive movement strategies**: break tasks into steps, rehearse mentally, perform each step consciously (sit to stand: "nose over toes, push up, stand tall").
- **Aerobic** exercise: 3 days a week, 30–40 min, 60–85% HRmax.

Rigidity

- **Slow rhythmic rotation** of the trunk and limbs.
- **Rhythmic initiation** (PNF): passive → active-assisted → active.
- Large reciprocal movements (arm swing, cycling); trunk extension and rotation.
- **Flexibility** for pectorals, hip flexors and hamstrings; prone lying if tolerated.
- Relaxation; warm-up before exercise; chest expansion exercises.

7 marks Physiotherapy for Parkinson's disease at Hoehn and Yahr stage III

Asked: S26 (2g)

Stage III: both sides affected, mild to moderate disability, **postural instability** present (falls begin), still physically independent.

Aims: prevent falls, keep gait and transfers independent, maintain fitness and posture, keep social roles.

- **Balance and fall prevention:** multimodal balance (reactive steps, weight shifts, reaching, turning, dual tasks when safe) 2–3 times a week, 16–30 hours over 5–10 weeks.
- **Gait training with cues:** floor stripes, metronome, "big step"; wide-arc turns; strategies for freezing.
- **Cognitive movement strategies** for sit to stand, turning in bed, getting up from the floor.
- **Aerobic:** treadmill, cycling, brisk walking, 3 days a week, 30–40 min, moderate to high intensity.
- **Resistance:** legs and trunk extensors, 1–2 days a week; power training.
- **Posture:** trunk extension and rotation, chest expansion, flexibility.
- **Home safety:** remove rugs, grab bars, raised toilet, good lighting, stripes at freezing spots.
- **Community classes** (dance, tai chi) and behaviour change for lifelong exercise.

Train in the "on" period; check lying and standing BP; guard with a gait belt.

5A(j) Anterior horn cell diseases: MND, SMA and poliomyelitis

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3 marks Types of motor neuron disease

Asked: S21 (1b), W24 (1f) (asked 2 times)

Type	Neurons	Features
ALS (commonest)	UMN + LMN	Wasting, fasciculations with spasticity, brisk reflexes, Babinski
Progressive bulbar palsy	Brainstem nuclei	Dysarthria, dysphagia, wasted fasciculating tongue; worst outlook
Progressive muscular atrophy	LMN only	Wasting, fasciculations, lost reflexes
Primary lateral sclerosis	UMN only	Slow spasticity; best outlook

3 marks Diagnostic criteria of motor neuron disease

Asked: S18 (1d)

Gold Coast criteria (2020):

1. **Progressive motor impairment** after previously normal function, shown by history or repeated examination.
2. **UMN and LMN** dysfunction in at least **one** body region, or LMN dysfunction in at least **two** regions.
3. **Investigations exclude** other diseases (MRI, blood tests).

Older **revised El Escorial** criteria graded ALS as definite, probable or possible by the number of regions (bulbar, cervical, thoracic, lumbosacral) with UMN and LMN signs. **EMG** (fibrillations, fasciculations) shows LMN involvement even in normal-looking muscles; sensory studies are normal.

3 marks Pseudobulbar palsy

Asked: S19 (1b)

Pseudobulbar palsy: UMN weakness of muscles supplied by cranial nerves IX–XII, from **bilateral corticobulbar tract** lesions (bilateral strokes, MND, MS).

Feature	Pseudobulbar (UMN)	Bulbar (LMN)
Tongue	Small, stiff; no wasting	Wasted, fasciculating
Speech	Spastic, slow, strained	Nasal, slurred
Jaw jerk	**Brisk**	Absent or normal; weak gag
Emotion	**Emotional lability**	Normal

Dysphagia brings aspiration risk: upright for meals, refer to speech therapy.

7 marks Classify motor neuron disease with features

Asked: S23 (2e)

MND: progressive degeneration of upper motor neurons (cortex) and lower motor neurons (brainstem, anterior horn),

with **sensation, eye movements and bladder spared**; onset usually after 50; 10% familial.

1. Amyotrophic lateral sclerosis (ALS, about 70%)
- UMN + LMN together: wasting, **fasciculations**, cramps with spasticity, brisk reflexes and Babinski in the same limb.
- Often starts in one hand ("split hand": thenar wasting more than hypothenar) or foot drop.

2. Progressive bulbar palsy
- Brainstem motor nuclei: slurred nasal speech, dysphagia, drooling, **wasted fasciculating tongue**; often pseudobulbar (UMN) features too. Worst outlook (aspiration, malnutrition).

3. Progressive muscular atrophy
- LMN only: wasting, fasciculations, weakness, lost reflexes; usually starts distally.

4. Primary lateral sclerosis
- UMN only: slowly progressive spasticity (legs first), brisk reflexes; best outlook.

Course: neck extensor weakness (head drop), breathing weakness (first at night), weight loss; median survival about 3 years. Function scale: ALSFRS-R (0–48).

Physiotherapy in one line: non-fatiguing exercise early; orthoses and aids in the middle stage; chest care, positioning and carer training late.

7 marks Physiotherapy for a terminally ill patient with MND

Asked: W25 (2b)

Aims: comfort, dignity, prevent painful complications, ease breathlessness and secretions, support the carers; follow the person's wishes and advance care plan with the palliative team.

- **Comfort and positioning:** supported positions with pillows, profiling bed, pressure-relieving mattress; regular turning.
- **Range of motion:** gentle passive movements to prevent painful stiff joints, especially shoulders.
- **Breathlessness:** upright or forward-lean positions; a **hand-held fan**; breathing control and relaxation; support **NIV** use and mask comfort.
- **Secretions:** positioning, gentle airway clearance, **manually assisted cough** or cough-assist machine, suction as needed.
- **Equipment:** hoist, slide sheets, tilt-in-space wheelchair, head support and neck collar.
- **Aspiration:** upright for meals and 30 minutes after.
- **Carers:** safe handling and positioning, pressure care; emotional support for the family.

No strengthening or fatiguing exercise at this stage; treatment is about comfort, not cure.

5A(k) Myopathies and myasthenia gravis

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3 marks Three features of myasthenia gravis

Asked: W24 (1a)

Autoimmune disease of the **neuromuscular junction**: antibodies block ACh receptors.

1. **Fatigable weakness:** worse with repeated use and towards evening, better after rest; proximal limbs and neck.
2. **Ocular: ptosis** and **diplopia** (commonest first symptom).
3. **Bulbar:** slurred or nasal speech, difficulty chewing and swallowing, "snarling" smile; breathing can fail (myasthenic crisis).

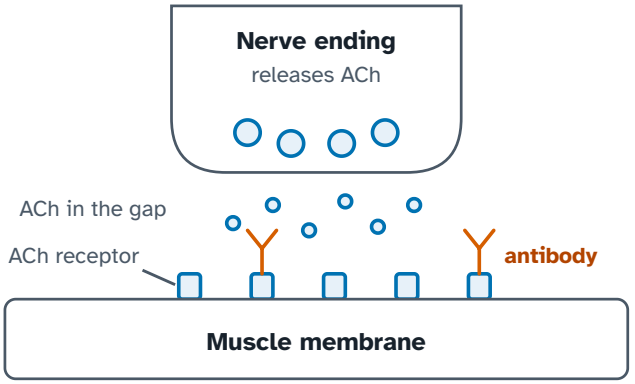
Reflexes, sensation and bladder are normal; wasting unusual.

7 marks Physiotherapy management of myasthenia gravis

Asked: W19 (2a)

MG: antibodies against ACh receptors at the neuromuscular junction cause **fatigable** weakness: ptosis, diplopia, bulbar, neck, proximal limb and breathing muscles.

What goes wrong at the neuromuscular junction



In MG, antibodies block and destroy receptors, so each signal opens fewer of them: weakness that worsens with repeated use.

Pyridostigmine slows the breakdown of ACh.

Draw this for the features question: it explains why the weakness fatigues and why pyridostigmine helps.

Aims: keep strength and function, protect breathing, manage fatigue, educate about crisis.

- **Exercise** (stable MG): moderate resistance and aerobic training, started low and built up; short sets with rests; aim for 150 min a week of activity; no exhaustion (Gilhus 2021).
- **Timing:** about **1–2 hours after pyridostigmine**, earlier in the day, in a **cool** room.
- **Breathing:** inspiratory muscle training (3 times a week for 8 weeks, 20% rising to 60% of MIP) with breathing retraining; huffing, assisted cough, positioning.

- **Fatigue:** energy conservation: plan, prioritise, pace; sit for tasks.
- **Aids:** soft collar for head drop; eyelid support or alternating eye patch; fall-proof home.
- **After thymectomy:** breathing exercises, early walking, sternal precautions.
- **Education:** triggers (infection, heat, stress, some drugs) and signs of crisis.

No exercise during a flare, infection or crisis; avoid heat; stop and call the team if breathing, speech or swallowing worsen.

5A(l) Disorders of the autonomic nervous system

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3 marks Horner's syndrome: features

Asked: S23 (1e)

Loss of the **sympathetic** supply to one side of the face and eye. Same-side features:

- **Ptosis** (mild drooping of the upper lid).
- **Miosis** (small pupil, most obvious in dim light; still reacts to light).
- **Anhidrosis** (no sweating of the face); apparent **enophthalmos**.

Causes by neuron: first order (lateral medullary stroke, cervical cord injury, syringomyelia); second order (**T1 root avulsion** in Klumpke's palsy, Pancoast tumour); third order (carotid dissection). In plexus injury it means preganglionic avulsion: poor outlook for the hand.

Part TP: Physiotherapy management: the adult treatment programme

TP2 Neurotherapeutic approaches: Brunnstrom, Rood, Bobath, MRP, PNF and CIMT

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3 marks Three Rood techniques to manage spasticity

Asked: S16 (1b)

1. **Slow stroking** down the back over the spinal muscles for 3–5 minutes (calming sensory input).
2. **Neutral warmth:** wrap the part or body in a blanket for 10–20 minutes.
3. **Prolonged (maintained) stretch** of the spastic muscle.

Others: maintained pressure over the tendon insertion; slow rhythmic rocking; light joint compression; prolonged cold. Effects are short-lived: follow at once with active practice.

3 marks Three principles of Rood's approach

Asked: S22 (1d)

1. **Normalise tone** and evoke the wanted response with the right **sensory stimulus**: some stimuli facilitate (fast brushing, quick icing, tapping), others inhibit (slow stroking, neutral warmth).
2. Sensorimotor control develops in a **developmental (ontogenetic) sequence**: supine withdrawal, roll over, pivot prone, neck co-contraction, prone on elbows, all fours, standing, walking; levels mobility → stability → controlled mobility → skill.
3. Movement is **purposeful** and needs **repetition** to be learned.

3 marks Three principles of Bobath (NDT)

Asked: W16 (1f)

1. **Inhibit** abnormal tone and abnormal postural reflexes, which block normal movement (reflex-inhibiting patterns).
2. **Facilitate** normal automatic postural reactions (righting, equilibrium, protective) and normal movement patterns through handling at **key points of control** (head, shoulder girdle, pelvis; hands and feet).
3. Treat the person as a whole and **include the affected side** in all activity (bilateral, symmetrical).

Modern Bobath is a problem-solving approach for postural control and selective movement; not superior to other approaches after stroke.

3 marks Reflex-inhibiting patterns with an example

Asked: W19 (1a)

Reflex-inhibiting patterns (Bobath): positions or movements **opposite to the abnormal spastic pattern**, used to reduce tone before facilitating movement.

- **Hemiplegic arm** (flexor pattern): shoulder forward and externally rotated, elbow extended, forearm supinated, wrist and fingers extended, thumb abducted; for example, sitting with weight on the extended arm beside or behind the body.
- **Hemiplegic leg** (extensor pattern): hip and knee flexed, hip abducted and externally rotated, ankle dorsiflexed.
- **Child with extensor thrust:** flexed position (head, trunk, hips, knees bent) in side lying or carried in flexion.

3 marks Three principles of the Motor Relearning Programme

Asked: S18 (1e)

1. **Task-specific practice:** recovery is relearning of real-life tasks, practised in the right context.
2. **Active learner:** the therapist teaches with instruction, demonstration and feedback; eliminate unnecessary muscle activity and practise the **missing components**.
3. **Practice and carry-over:** many repetitions, continued on the ward and at home.

Four steps: analyse the task → practise missing components → practise the whole task → transfer to daily life (Carr and Shepherd).

7 marks Principles and techniques of Rood's approach

Asked: S17 (2d)

Principles

- Normalise tone and evoke the wanted motor response with appropriate **sensory stimulation**.
- Follow the **ontogenetic motor sequence**: supine withdrawal → roll over → pivot prone → neck co-contraction → prone on elbows → all fours → standing → walking.
- Levels of control: **mobility, stability, controlled mobility, skill**.
- Purposeful activity calls out the response automatically; repetition is needed for learning.

Facilitatory techniques (hypotonia)

- **Fast brushing** over the skin of the muscle.
- **Quick icing** (brief strokes of ice).
- **Quick stretch, tapping** over the muscle belly or tendon.
- **High-frequency vibration**.
- **Heavy joint compression** for co-contraction; resistance.
- **Fast vestibular** stimulation (fast rocking, spinning).

Inhibitory techniques (spasticity)

- **Slow stroking** down the back for 3–5 minutes.
- **Neutral warmth** (blanket wrap) for 10–20 minutes.
- **Prolonged stretch**; **maintained tendon pressure**.
- Light joint compression; slow rhythmic rocking; slow vestibular input; prolonged cold.

Effects are short-lived and evidence is limited: use them to prepare for active, task-specific practice (Cochrane 2014: no approach is superior).

7 marks Principles of proprioceptive neuromuscular facilitation (PNF)

Asked: W25 (2c)

Philosophy: a positive approach that builds on what the person can do, works on function, uses strong parts to help the weak, treats the whole person and uses motor learning.

Basic principles (procedures)

- **Resistance:** optimal, to facilitate contraction and learning.
- **Irradiation and reinforcement:** strong muscles spread activity to weak ones in the pattern.
- **Manual contact:** lumbrical grip over the working muscles guides direction.
- **Body position and mechanics** of the therapist in line with the movement.
- **Verbal commands:** short, clear, well timed; **vision**: the patient watches.
- **Traction** (facilitates movement) and **approximation** (facilitates stability).
- **Stretch** stimulus and stretch reflex.
- **Timing:** normal distal-to-proximal; timing for emphasis.
- **Patterns:** diagonal, spiral mass movements (D1 and D2 flexion and extension).

Neurophysiological basis (Sherrington): after-discharge, temporal and spatial summation, irradiation, successive induction, reciprocal innervation.

Techniques: rhythmic initiation, combination of isotonics, repeated stretch; dynamic and stabilising reversals, rhythmic stabilisation; contract-relax, hold-relax.

> **Draw:** Upper limb D2 flexion ("drawing a sword") and extension ("sheathing the sword").

TP2 Managing tone: spasticity, rigidity and hypotonia

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3 marks Define and classify muscle tone

Asked: S20 (1e)

Muscle tone: the resistance of a relaxed muscle to passive stretch, with a **neural** part (stretch reflex, descending control) and a **non-neural** part (muscle and connective tissue stiffness).

- **Hypertonia: spasticity** (velocity-dependent, clasp-knife; UMN); **rigidity** (lead-pipe or cogwheel, any speed; basal ganglia); **dystonia**.
- **Hypotonia (flaccidity):** low resistance, floppy, hypermobile (LMN, cerebellum, spinal shock, Down's).
- **Fluctuating tone:** swings low to high (athetoid CP).

3 marks Slow icing

Asked: W18 (1b)

- **Method:** an ice pack wrapped in a damp towel, or cold immersion, over the spastic muscle for about **15–20 minutes** (prolonged cold).
- **Effect:** cooling slows nerve conduction and reduces muscle spindle activity, so **tone drops for a short time**.
- **Use:** the low-tone window for stretching and active practice (spastic hand, calf).
- **Precautions:** check sensation and circulation; avoid in cold sensitivity, Raynaud's, numb skin.

Contrast: quick icing (brief strokes) facilitates.

3 marks Three facilitatory techniques for hypotonia

Asked: W16 (1d)

1. **Quick stretch and tapping** over the muscle belly or tendon before movement: excites the muscle spindle.
2. **Approximation** (joint compression) in weight-bearing positions: joint receptors facilitate co-contraction.
3. **Fast brushing or quick icing** over the skin of the muscle (Rood).

Others: vibration, resistance, bouncing on a ball, fast vestibular input, PNF repeated stretch.

3 marks Three methods to reduce tone

Asked: S23 (1b)

1. **Prolonged stretch** by positioning, splint, standing frame or serial cast: reduces reflex excitability and keeps length.
2. **Prolonged cold (slow icing)** for 15–20 minutes: slows conduction and spindle activity.
3. **Weight-bearing** through the extended arm or a plantigrade foot: stretch plus compression.

Others: neutral warmth, slow rhythmic rotation, reflex-inhibiting patterns, electrical stimulation of antagonists.

7 marks Muscle tone; physiotherapy of hypotonia (3 + 4)

Asked: W17 (2d)

Muscle tone (3)

The resistance of a relaxed muscle to passive stretch: neural (stretch reflex, descending control) plus non-neural (tissue stiffness). Normal tone is high enough to hold posture against gravity and low enough for selective movement. Types: spasticity, rigidity, dystonia, hypotonia, fluctuating tone.

Physiotherapy of hypotonia (4)

- **Protect** hypermobile joints (shoulder support in early stroke; no overstretching).
- **Positioning against gravity** with support: prone over a wedge, supported sitting, standing frame.
- **Facilitation** just before active movement: quick stretch, tapping, fast brushing, quick icing, vibration, fast vestibular input.
- **Co-contraction:** approximation and weight-bearing in kneeling and standing; rhythmic stabilisation.
- **Strengthening and function:** head and trunk control, sit to stand, walking; electrical stimulation of weak muscles.
- Teach carers handling and safe positions.

7 marks Facilitatory techniques to improve tone

Asked: W24 (2d)

Technique	How	Why it works
Quick stretch, tapping	Brisk stretch or tap over belly or tendon before movement	Excites spindle and stretch reflex
Approximation	Compression through the long axis in weight-bearing	Joint receptors facilitate co-contraction
Resistance	Light, graded resistance to movement or holding	Recruits motor units; irradiation
Fast brushing, quick icing	Brief stimulation over the skin of the muscle	Facilitatory input (Rood)
Vibration	High-frequency over the muscle belly	Tonic vibration reflex
Weight-bearing, bouncing	Prone on elbows, four-point, ball bouncing, standing	Compression and load changes activate postural muscles
Fast vestibular input	Brisk rocking or swinging	Raises postural tone
PNF	Repeated stretch, rhythmic stabilisation, stabilising reversals	Strength and stability
Electrical stimulation	Weak muscles during function	Contraction while control returns

Precautions: quick stimuli can raise spasticity elsewhere; avoid fast vestibular input in seizures or raised ICP. Follow facilitation with active task practice.

7 marks Tone-inhibiting techniques; physiotherapy of spasticity

Asked: S19 (2b), W20 (2a) (asked 2 times)

Spasticity (Lance 1980): velocity-dependent increase in tonic stretch reflexes with exaggerated tendon jerks, part of the UMN syndrome. Measure with MAS and Tardieu.

First: treat only if it causes a problem (pain, contracture, hygiene, function); look for **triggers** (bladder, constipation, pressure sore, ingrown toenail, tight clothes, anxiety).

Technique	How and why
Positioning	Opposite to the spastic pattern (reflex-inhibiting patterns) in lying, sitting, standing
Prolonged stretch	Held for minutes by positioning, splint, standing frame or cast
Weight-bearing	Extended arm in sitting; plantigrade foot in standing
Slow icing	15–20 minutes; tone drops for a short time
Neutral warmth, slow stroking	Calming input (Rood)
Rhythmic rotation	Slow rotation of trunk or limb through increasing range
Splints and serial casting	Low-load, long-duration stretch
Electrical stimulation	Antagonists during task practice (reciprocal inhibition)
Active task practice	Strengthen weak antagonists; reach, walk

Team: baclofen, tizanidine; **botulinum toxin** for focal spasticity followed by stretching, casting and practice; intrathecal baclofen.

Never force a spastic joint. Passive stretch alone has little effect on spasticity (Cochrane); combine with active practice.

TP2 Balance and coordination training

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3 marks Three motor strategies of balance

Asked: S18 (1a)

Strategy	When	How
Ankle	Small, slow disturbances; firm, wide surface	Body sways as one piece about the ankles; muscles fire distal to proximal
Hip	Larger, faster disturbances; narrow or soft surface	Quick hip flexion or extension moves the centre of mass; proximal to distal
Stepping (reaching)	Centre of mass moves outside the base	A rapid step or reach creates a new base of support

A suspensory strategy (bending the knees to lower the centre of mass) is sometimes added.

3 marks Three strategies to improve dynamic balance

Asked: S17 (1a)

1. **Reaching** beyond arm's length in all directions, to the floor and overhead.
2. **Weight shifts and stepping:** forward, backward, sideways, onto a step, over obstacles; turning.
3. **Reactive (perturbation) training:** expected, then unexpected pushes or pulls to train ankle, hip and stepping responses.

Also: walking variations (speed, direction, head turns, dual tasks); tai chi, dance, ball games. Dose: more than 3 hours a week of challenging balance exercise reduces falls by 39%.

7 marks Training static and dynamic standing balance

Asked: S16 (2b)

Balance keeps the centre of mass over the base of support: static, dynamic and reactive. It needs vision, vestibular and somatosensory input, central integration, strength and range, and attention.

Progression (one variable at a time)

Change	Easier → harder
Base	Feet apart → together → semi-tandem → tandem → single leg
Surface	Firm → foam → wobble board → uneven ground
Vision	Eyes open → closed → moving visual scenes
Arm support	Parallel bars → no hands → holding an object
Head	Still → turning and nodding
Attention	Single → dual task (talking, counting, carrying a tray)

Static balance

Hold each position (for example 30 seconds) before progressing; equal weight on both legs with mirror or scale feedback.

Dynamic balance

- Reaching in all directions; weight shifts; stepping and turning; step-ups; obstacles.
- Perturbation training for ankle, hip and stepping responses.
- Walking with speed and direction changes, head turns, dual tasks; tai chi, dance, ball games.

Dose: challenging balance exercise >3 hours a week (Sherrington 2017). **Safety:** gait belt, parallel bars, stand on the weaker side, clear the area.

7 marks Evaluating sensory and motor strategies of balance

Asked: W18 (2b)

Sensory strategies

- **mCTSIB:** 30 seconds in four conditions: firm eyes open, firm eyes closed, foam eyes open, foam eyes closed. Falls on firm surface with eyes closed → somatosensory loss; falls on foam with eyes closed → vestibular loss or poor sensory reweighting.
- **Romberg and sharpened Romberg:** loss of balance on eye closure → sensory (proprioceptive) ataxia.
- **Sensory Organisation Test** on posturography (six conditions) where available.

Motor strategies

- **Reactive responses:** push-and-release test or nudges in each direction; observe ankle, hip or stepping response and its speed. Pull test in Parkinson's.
- **Limits of stability and anticipatory control:** functional reach (<15 cm = high fall risk); multidirectional reach; rising on toes.
- **Systems tools:** BESTest or Mini-BESTest (biomechanical constraints, stability limits, anticipatory, reactive, sensory orientation, dynamic gait).

Function and fall risk

Berg Balance Scale (0–56), Timed Up and Go, Dynamic Gait Index or Functional Gait Assessment, dual-task walking.

TP5 Bladder and bowel training

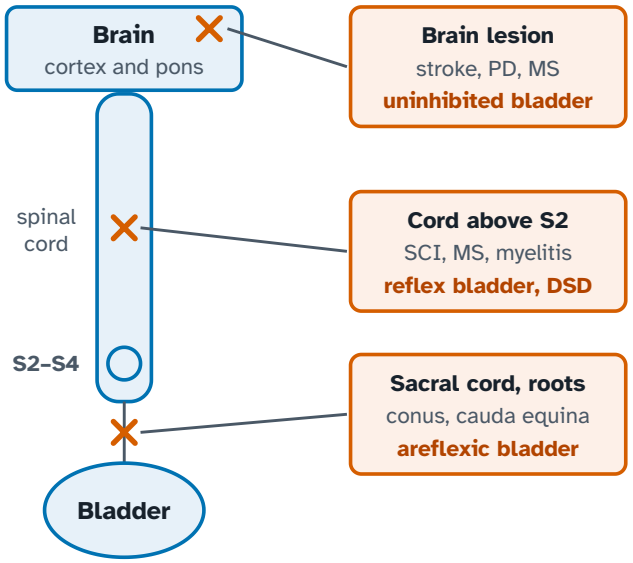
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3 marks Types of neurological (pathological) bladder

Asked: S17 (1d), W20 (1b), S23 (1d) (asked 3 times)

- **Uninhibited:** brain lesion (stroke, PD, MS); urgency, frequency, urge leaks; empties fully.
- **Reflex (automatic, spastic):** complete cord lesion above S2; reflex emptying without sensation; dyssynergia.
- **Autonomous (atonic, flaccid):** sacral cord or cauda equina; large capacity, overflow dribbling.
- **Sensory paralytic:** diabetes, tabes, B12 deficiency; no sense of fullness.
- **Motor paralytic:** polio, root injury; sensation present but cannot empty.

Where the lesion is decides the type of bladder



Crosses mark the three lesion sites. DSD is detrusor-sphincter dyssynergia: the sphincter tightens when the bladder contracts.

3 marks Reflex bladder

Asked: W22 (1b)

- Follows a **complete cord lesion above the sacral segments** (S2–S4).
- The sacral reflex arc is **intact but cut off from the brain**, so the bladder contracts by reflex when it fills, without sensation or voluntary control; small capacity.
- The sphincter often tightens at the same time (**detrusor-sphincter dyssynergia**): high pressure, incomplete emptying, risk of reflux, kidney damage and autonomic dysreflexia (T6 and above).
- **Management:** clean intermittent catheterisation with drugs that calm the detrusor.

7 marks Spastic vs flaccid bladder and their physiotherapy (3 + 4)

Asked: W16 (2b), W17 (2f) (asked 2 times)

Feature	Spastic (UMN, reflex)	Flaccid (LMN, areflexic)
Lesion	Cord above S2–S4 (fracture above about T12)	Sacral cord or cauda equina (T12–L1 or below); spinal shock
Reflex arc	Intact, cut off from brain	Broken
Detrusor, capacity	Overactive; small	Underactive; large
Sphincter	Dyssynergia	Weak (stress leaks) or does not open
Symptoms	Sudden reflex leaks, incomplete emptying	Retention with overflow dribbling
Danger	High pressure, kidney damage, autonomic dysreflexia	Overstretching, infection, stones

Physiotherapy

- **Both:** teach **clean intermittent catheterisation** 4–6 times a day, volumes under 400–500 mL: hand washing, long sitting or wheelchair position, sitting balance with both hands free, clothing; adapted aids for weak hands.
- **Spastic:** plus antimuscarinic drugs; teach autonomic dysreflexia signs and first steps.
- **Flaccid:** pelvic floor training if any contraction; protect skin from leaks.
- Transfers to toilet or commode; bladder diary; fluid plan; watch for infection.

Suprapubic tapping (spastic) and Credé or Valsalva (flaccid) raise bladder pressure: only if urodynamics show safe pressures (EAU 2026).

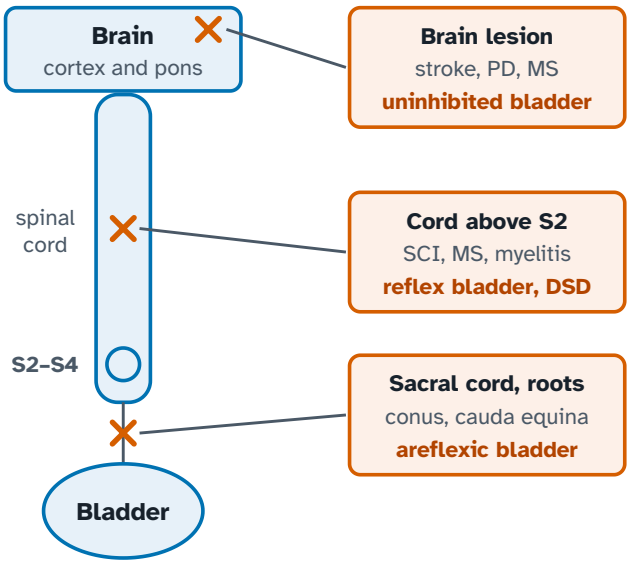
7 marks Classify neurogenic bladder; assessment and physiotherapy management

Asked: S21 (2d), W25 (2a), S26 (2c) (asked 3 times)

Classification

- **By lesion level (EAU):** above the pons (detrusor overactivity, sphincter normal); spinal cord between pons and sacral cord (overactivity with dyssynergia); sacral cord and below (weak or absent detrusor).
- **Classic (Lapides):** uninhibited, reflex, autonomous, sensory paralytic, motor paralytic.

Where the lesion is decides the type of bladder



Crosses mark the three lesion sites. DSD is detrusor-sphincter dyssynergia: the sphincter tightens when the bladder contracts.

Assessment

- How the person empties; leaks, urgency, infections, autonomic dysreflexia, bowel pattern.
- **Bladder diary** for 3 days; **residual urine** (scanner); urodynamics by the urologist.
- Hand function, sitting balance, transfers, dressing, cognition; pelvic floor contraction; home toilet type.

Management by type

Type	Physiotherapy
Uninhibited	**Timed voiding**; **urge suppression** (stop, squeeze pelvic floor, breathe, walk calmly); pelvic floor training ≥3 months; tibial nerve stimulation; faster transfers; bedside commode; fluids and caffeine advice
Reflex	**CIC** 4–6 times a day (<400–500 mL) with antimuscarinic drugs; self-catheterisation skills; autonomic dysreflexia education
Areflexic	CIC; pelvic floor training if any contraction; skin protection
Sensory paralytic	Void by the clock; catheterise if residual stays high

Tapping, Credé and Valsalva only if urodynamics show safe pressure; never block catheter drainage on the tilt table.

TP7 Orthoses and splints

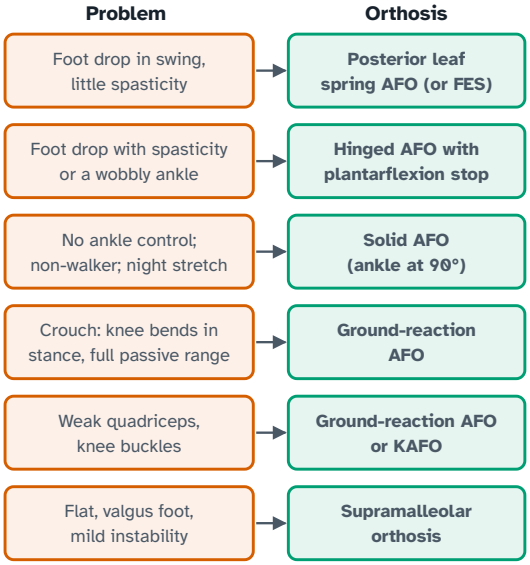
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3 marks Ankle-foot orthosis

Asked: W21 (1b)

- **Parts:** calf section with strap, ankle section (solid or hinged), footplate; worn in a shoe.
- **Principle: three-point force system** controls the ankle (footplate lifts the forefoot, strap over the front of the ankle, calf shell behind); through the ground reaction force it also controls the knee.
- **Uses:** foot drop (stroke, peroneal injury, MS), equinus and spasticity, ankle instability, crouch gait, knee hyperextension.
- **Types:** posterior leaf spring, solid, hinged with plantarflexion stop, ground-reaction, supramalleolar.

Choosing an ankle-foot orthosis



A ground-reaction AFO needs a knee and hip that can straighten fully; if they cannot, or the hip is weak too, use a KAFO.

▶ Text version of the chart

3 marks Orthosis used for radial nerve injury

Asked: W17 (1f)

Radial palsy gives **wrist, finger and thumb drop**, so grip is weak.

- **Static cock-up splint** at rest and night: holds the wrist in 20–30° extension, ending at the distal palmar crease so the fingers move freely; prevents overstretching of the extensors.
- **Dynamic cock-up (lively) splint** by day: wrist support plus an **outrigger with elastic slings** under the proximal phalanges and thumb; the patient flexes actively to grip and the elastics re-extend the fingers, so grasp and release are possible.

Draw: A forearm splint with the wrist extended and an outrigger over the knuckles with finger slings.

3 marks Dynamic cock-up splint: uses and principle

Asked: S25 (1b)

- **Uses:** radial nerve palsy (wrist, finger and thumb drop); after extensor tendon repair; after tendon transfers for radial palsy.
- **Principle:** the splint **replaces the paralysed extensors**. A static wrist support keeps the wrist extended so the flexors grip strongly (tenodesis advantage); **elastic slings on an outrigger** return the fingers and thumb to extension after each active grasp.
- **Benefits:** allows function (grasp and release), prevents overstretching of extensors, stops the flexors shortening.

3 marks Cock-up splint

Asked: W25 (1f)

- **Description:** a forearm-based (volar) static splint holding the wrist in about **20–30° of extension** (neutral for carpal tunnel), from two-thirds of the forearm to the **distal palmar crease**, clear of the thumb, so the fingers and thumb move freely; straps at forearm, wrist and palm.
- **Uses:** radial palsy (rest and night); **carpal tunnel syndrome** (neutral, at night); wrist fractures, arthritis, spastic wrist.
- **Making:** paper pattern → low-temperature thermoplastic heated at 65–70 °C → mould over stockinette → trim, pad, straps → check fit.
- **Care:** check the skin; keep away from heat.

Part 5B: Physiotherapy management: paediatric conditions

5B Developmental neurology: milestones, reflexes and reactions

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3 marks Asymmetrical tonic neck reflex (ATNR)

Asked: S20 (1a), S21 (1f) (asked 2 times)

- **Elicit:** child supine, head in midline; turn the head to one side.
- **Response:** arm (and leg) on the **face side extend**, those on the **skull side flex**: "fencing" posture.
- **Normal:** birth to about **6 months**; then integrated so hands can come to midline and the child can roll.
- **Abnormal:** obligatory at any age or persisting beyond 6 months (CP): cannot bring hands together or to mouth, cannot roll or sit symmetrically, poor hand–eye coordination, asymmetry with risk of scoliosis and hip problems.
- **PT:** midline positioning (side lying, symmetrical seating, toys in front).

3 marks Any one primitive reflex: the Moro reflex

Asked: W20 (1e)

- **Elicit:** let the supported head drop back suddenly a few centimetres.
- **Response:** arms **abduct and extend** with hands opening, then **adduct and flex** (embrace), often with a cry.
- **Normal:** birth to 3–4 months (up to 6).
- **Abnormal:** absent (severe brain depression, hypotonia); **asymmetrical** (brachial plexus palsy, fractured clavicle or humerus, hemiplegia); persisting beyond 6 months (CP).

3 marks Cortical reflexes

Asked: W20 (1f)

Highest level of the classic reflex hierarchy (Fiorentino): **equilibrium (tilting) reactions**, which appear once the child can get into each position and persist for life.

- Equilibrium reactions in **prone** and **supine** (about 6 months).
- In **four-point kneeling** and **sitting** (about 7–8 months).
- In **kneel-standing** and **standing** (second year).
- **Hopping** and **dorsiflexion** reactions in standing.

Response to a tilt: trunk curves towards the higher side; limbs on the higher side abduct and extend. The hierarchy is a teaching model; skills are trained directly.

3 marks Parachute reactions

Asked: W25 (1d)

- **Elicit:** hold the child prone around the trunk and move suddenly head-first towards a surface.
- **Response:** both arms **extend and abduct**, hands open, to protect the head.
- **Timing:** forward at about **6–9 months**, then sideways and backwards; persists for life.
- **Abnormal:** absent, delayed or asymmetrical (hemiplegic CP, brachial plexus palsy, motor delay): sitting and standing are unsafe.
- **PT:** train on a ball, roll or tilt board.

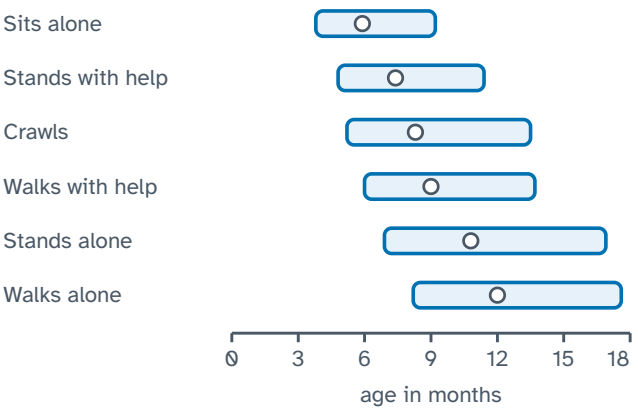
7 marks Gross and fine motor milestones up to 5 years

Asked: W19 (2e), S21 (2c), S23 (2b) (asked 3 times)

Principles: sequence is fixed, rate varies; **cephalocaudal** (head before feet), **proximodistal** (trunk before fingers), general to specific; use corrected age for preterm babies until 2 years.

Age	Gross motor	Fine motor
3 months	Head steady; chest up on forearms in prone	Watches hands; hands to midline
4–5 months	Rolls over (5 months)	Bidextrous reach (4 months)
6 months	Sits with support; weight on straight arms in prone	Unidextrous reach; palmar grasp; transfers by 6–7 months
8–9 months	Sits without support (8); stands holding on, crawls (9)	Immature pincer (9)
12 months	Stands alone; walks with one hand held	Mature pincer grasp
15 months	Walks alone; creeps upstairs	Scribbles in imitation; tower of 2
18 months	Runs stiffly; climbs onto furniture	Scribbles; tower of 3; spoon feeding
2 years	Runs well; upstairs two feet per step; jumps; kicks a ball	Tower of 6; vertical and circular strokes
3 years	Upstairs with alternate feet; rides a tricycle	Tower of 9; copies a circle; dresses with help
4 years	Hops on one foot; downstairs alternate feet	Copies a cross; builds a bridge
5 years	Skips; hops well on either foot	Copies a triangle; ties shoelaces

Normal age range for six gross motor milestones



Bars: the normal range (1st to 99th percentile) in 816 healthy children from Ghana, India, Norway, Oman and the USA; dots: the median age (WHO Motor Development Study, 2006). Walking alone anywhere between about 8 and 18 months is normal.

Normal ranges are wide: walking alone anywhere between about 8 and 18 months is normal (WHO 2006). Red flags: loss of skills, asymmetry, hand preference before 12 months.

5B(1) Cerebral palsy

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3 marks Causes (aetiological factors) of cerebral palsy; prenatal causes

Asked: W16 (1b), S18 (1c), W17 (1c), W19 (1b), W22 (1c) (asked 5 times)

Period	Causes
Prenatal	**Maternal infections** (rubella, CMV, toxoplasmosis); **multiple pregnancy**; **placental insufficiency** and poor fetal growth; pre-eclampsia, maternal thyroid disease, epilepsy; brain malformations; alcohol, drugs; Rh incompatibility
Perinatal	**Prematurity** (<32 weeks) and low birth weight; **birth asphyxia** (delayed cry, HIE); prolonged labour; **kernicterus**; neonatal infection or stroke
Postnatal (first 2 years)	**Meningitis**, encephalitis; head injury; near drowning; severe hypoglycaemia

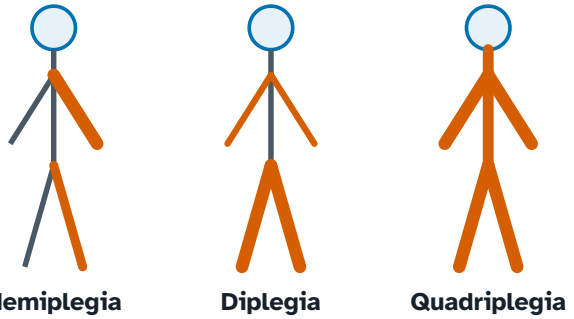
For "three prenatal causes", give three from the first row only.

3 marks Classification of cerebral palsy

Asked: S26 (1e)

- **By movement disorder:** spastic (commonest), dyskinetic (athetoid, dystonic), ataxic, hypotonic, mixed.
- **By distribution** (spastic): hemiplegia (arm more than leg), diplegia (legs much more than arms), quadriplegia (all four limbs and trunk).
- **By function:** GMFCS levels I–V (walks without limitation → transported in a wheelchair); MACS I–V for hand use; CFCS for communication.

Distribution of spastic cerebral palsy



Thick orange lines show the affected parts; thicker means more affected. Hemiplegia: arm more than leg. Diplegia: legs much more than arms. Quadriplegia: all four limbs and the trunk.

3 marks Three impairments in hypotonic cerebral palsy

Asked: S26 (1b)

1. **Low tone** with weak antigravity muscles: poor **head and trunk control**; "frog-leg" lying posture.
2. **Hypermobile joints** and poor co-contraction (unstable proximal joints).
3. **Very delayed milestones**; low endurance.

Also: shallow breathing and weak cough; feeding difficulty. Often changes later to dyskinetic or ataxic CP.

3 marks Three spinal deviations in quadriplegic cerebral palsy

Asked: W25 (1a)

1. **Neuromuscular scoliosis:** long C-shaped curve, often with **pelvic obliquity**.
2. **Kyphosis** from poor trunk control in sitting.
3. **Increased lumbar lordosis** with hip flexion contracture.

Often with **windswept hips** and hip displacement. Prevention: supportive seating, positioning, standing programme, hip and spine surveillance.

7 marks Classification and aetiology of cerebral palsy

Asked: S24 (2e)

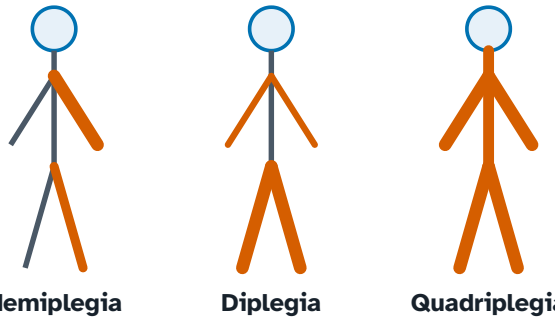
CP: a group of permanent disorders of movement and posture, causing activity limitation, due to a **non-progressive** disturbance of the developing fetal or infant brain (Rosenbaum 2007); about 3 per 1000 in India.

Classification

Type	Features	Lesion
Spastic	Velocity-dependent high tone, brisk reflexes, weakness; hemiplegia, diplegia, quadriplegia	Diplegia: periventricular leukomalacia (preterm); hemiplegia: one-sided stroke
Dyskinetic	Athetosis, chorea, dystonia; fluctuating tone	Basal ganglia: asphyxia at term, kernicterus
Ataxic	Low tone, poor balance, wide base, intention tremor	Cerebellum
Hypotonic	Floppy, very delayed milestones	Variable
Mixed	Spastic with dyskinetic features	Several areas

Functional: GMFCS I–V; MACS; CFCS.

Distribution of spastic cerebral palsy



Thick orange lines show the affected parts; thicker means more affected. Hemiplegia: arm more than leg. Diplegia: legs much more than arms. Quadriplegia: all four limbs and the trunk.

Aetiology

- **Prenatal:** maternal infection (TORCH), multiple pregnancy, placental insufficiency, pre-eclampsia, malformations, Rh incompatibility.
- **Perinatal:** prematurity and low birth weight, birth asphyxia, prolonged labour, kernicterus, neonatal infection.
- **Postnatal:** meningitis, encephalitis, head injury, near drowning, hypoglycaemia.

7 marks Posture and movement problems of a child with spastic diplegic CP

Asked: S26 (2b)

Spastic diplegia: legs much more affected than arms; often preterm with **periventricular leukomalacia**; intelligence often normal.

Posture

- Hips **flexed, adducted and internally rotated**; knees flexed; ankles in **equinus**.
- **Sitting:** on the sacrum with posterior pelvic tilt and rounded back (tight hamstrings), or **W-sitting** for stability.

Movement

- Poor selective control of the legs; moves legs together (commando crawl, bunny hopping).
- **Standing and walking: scissoring**, toe walking, **crouch** or jump-knee pattern, in-toeing (femoral anteversion); arms in high guard.
- Poor standing balance; associated reactions in the arms with effort.

Secondary problems

- Contractures of calf, hamstrings, hip flexors and adductors with growth; hip migration; torsional deformities.

Management in one line: stretching through casting and AFOs, strengthening, gait training with posterior walker and treadmill, botulinum toxin when indicated.

7 marks Posture and movement problems of a child with athetoid CP

Asked: S26 (2a)

Athetoid (dyskinetic) CP: basal ganglia damage from asphyxia at term or **kernicterus**.

Problems

- **Involuntary writhing movements** that increase with effort, excitement and attempts to move.
- **Fluctuating tone** from low to high.
- Poor **head and midline control**; persistent **ATNR** pulls the body into asymmetry on head turning.
- Poor **proximal stability** (little co-contraction): cannot hold positions, grades movement poorly; sudden **extensor thrusts** push the child out of the chair.
- Drooling, unclear speech, feeding difficulty; hearing loss after kernicterus; intelligence often normal.
- Walking, if achieved, is late, unsteady, wide-based with arms held high.

Physiotherapy approach

- Proximal stability: weight-bearing (prone on elbows, four-point, kneeling, standing frame); approximation; holding against gentle resistance.
- Midline control: head and hands in midline, symmetrical sitting, two-handed play.
- Graded, slow, small-range movement; avoid excitement.
- Stable seating with pelvic belt and foot support; walker or gait trainer when ready.

7 marks Carrying positions for children with CP (with rationale)

Asked: S18 (2a)

Child	How to carry	Rationale
Spastic, legs stiff and crossed	Facing you, legs apart around your waist or astride your hip, arms forward over your shoulders	Hip flexion and abduction break the extensor and **scissoring** pattern
Arches backwards	Curled forward: head, shoulders, hips and knees bent, arms forward	**Flexion** reduces extensor thrust
Athetoid	Arms and legs together and bent, close to your body; hands together in front; head in midline	Stability and **midline** control; limits involuntary movement
Floppy (hypotonic)	Back against your chest facing outwards; your arm under the hips and across the trunk; head supported	Supports the trunk; child can look around and lift the head
Hemiplegic	Affected arm forward; affected side lengthened over your arm	Counters the shortened, flexed side

General rules: give only as much support as needed so the child uses its own control; change positions; teach every family member.

7 marks Physiotherapy for spastic cerebral palsy

Asked: S19 (2e)

Principles: family-centred, goal-directed, active task-specific practice through play; start early; home programme; prevent deformity (Novak 2019).

- **Assessment:** GMFCS, GMFM, MAS and Tardieu, range (popliteal angle, Thomas, calf), selective control, gait, hips.
- **Range and contracture:** active use, **standing frame** in hip abduction (60 min a day), **serial casting**, AFOs; botulinum toxin plus stretching and practice.
- **Strength:** progressive training (sit to stand, step-ups), 2–4 times a week, 1–3 sets of 6–15, for 12–16 weeks.
- **Gait:** standing balance and weight shift; parallel bars → **posterior walker**; **treadmill** with or without body-weight support; AFO for equinus, ground-reaction AFO for crouch.
- **Hand (hemiplegia): CIMT** or bimanual training as play, about 40 hours in total.
- **Fitness:** cycling, swimming, active play ≥20 min, 2–3 times a week.
- **Positioning and seating**; avoid W-sitting; **hip surveillance** (refer if migration >30%).
- **Home programme** and parent training; school adaptations.

Passive NDT does not improve motor function; handling only sets the child up for active movement.

7 marks Foot orthoses used for children with cerebral palsy

Asked: S22 (2f)

Aims: hold the foot in alignment, prevent contracture, give a stable base for standing and improve gait efficiency.

Orthosis	What it does	Used for
Solid AFO	Holds the ankle at 90° in stance and swing	Equinus with poor control; standing in non-walkers; night resting splint
Hinged AFO with plantarflexion stop	Allows dorsiflexion, blocks plantarflexion	Walkers with enough calf length; more normal roll-over
Ground-reaction AFO	Anterior shell pushes the knee back in stance	**Crouch gait** (with good knee and hip range)
Posterior leaf spring	Flexible; lifts the foot in swing	Mild foot drop, little spasticity
Supramalleolar orthosis	Controls subtalar joint and forefoot	Flat, valgus feet, mild instability
Dynamic AFO	Thin, contoured footplate supports arches and toes	Mild tone problems; standing balance

Check skin daily; wear time builds up gradually; review with growth; combine with stretching and strengthening.

5B(2) Down's syndrome

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7 marks Physical traits and neurological impairments in Down's syndrome

Asked: S23 (2c)

Down's syndrome: trisomy 21 (non-disjunction 96%, translocation 3–4%, mosaicism 1–2%); risk rises with maternal age; about 1 in 1000 births in India.

Physical traits

- **Head and face:** small, flat-backed (brachycephalic) head; flat facial profile and nasal bridge; **upward-slanting eyes** with epicanthic folds; Brushfield spots; small mouth with **protruding tongue**; small low-set ears; excess skin at the back of the neck.
- **Hands:** short, broad; **single palmar crease**; clinodactyly of the little finger.
- **Feet:** wide **sandal gap** between first and second toes; flat feet.
- **Body:** short stature, hypotonia, hypermobile joints.

Neurological impairments

- **Hypotonia** and reduced strength.
- **Intellectual disability** (usually moderate, IQ 35–50); speech and language delay.
- **Delayed milestones:** sitting about 10 months, crawling 14 months, walking about 26 months.
- Poor balance and postural control; slow reaction times; poor motor planning.
- Gait: wide base, out-turned feet, knees locked in extension.
- Hearing loss (75%), vision problems, seizures, early Alzheimer's in adults.

Associated: congenital heart disease (40–50%), sleep apnoea, thyroid disease, atlantoaxial instability (1–2% symptomatic).

7 marks Physiotherapy management of a child with Down's syndrome

Asked: S20 (2d), W25 (2f) (asked 2 times)

Problems: hypotonia, lax joints, weak muscles, delayed milestones, poor balance, low endurance (heart), intellectual disability.

Assess: milestones, tone, laxity, balance, endurance, heart status, neck symptoms; AIMS, GMFM or Peabody.

- **Developmental training** in milestone order: head and trunk control (tummy time, prone play), sitting, crawling, pull to stand, cruising, walking; handling in midline; avoid the frog position.
- **Treadmill training:** supported stepping on a small treadmill, **8 minutes a day, 5 days a week**, from sitting alone; walked about **101 days earlier** (Ulrich 2001).
- **Strength and postural control:** weight-bearing play, squat to stand, step-ups, ball activities; progressive resistance in adolescents (twice a week for 10 weeks).
- **Balance** on varied surfaces; running, jumping, ball skills through play.
- **Fitness:** active play, swimming, cycling, dance; weight control.
- **Joint protection:** no end-range stretching; supramalleolar orthoses or insoles for flat feet.
- **Teaching style:** short, frequent, play-based sessions; demonstration, simple words, praise; parents run a daily home programme.

Safety: atlantoaxial instability signs (neck pain, head tilt, new weakness, gait or bladder change) → stop and refer; avoid trampolines; know the heart status.

5B(3) Neural tube defects: spina bifida and hydrocephalus

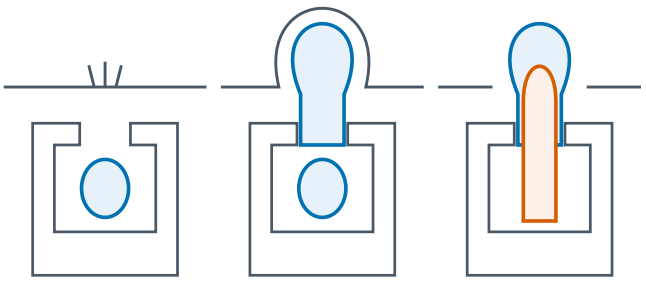
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3 marks Types of spina bifida

Asked: S25 (1e)

- **Spina bifida occulta:** gap in the bony arch only; skin closed (tuft of hair, dimple, fatty lump); usually no deficit.
- **Meningocele:** skin-covered sac of meninges and CSF; cord stays in the canal; few deficits.
- **Myelomeningocele:** cord and roots in the sac (often open); paralysis, sensory loss and neurogenic bladder below the level; commonest serious type.
- Others: lipomyelomeningocele, myeloschisis.

Types of spina bifida



Occulta bony gap only; skin closed

Meningocele meninges + CSF; cord in place

Myelomeningocele cord and roots in an open sac

Draw this for the types question. Blue is the sac of meninges and CSF (and the normal cord); orange is the cord pulled into the sac, which is why the nerves below it fail.

3 marks Deformities in spina bifida

Asked: S17 (1e)

- **Spine:** scoliosis, congenital kyphosis, lordosis.

- **Hip:** flexion contracture; flexion–adduction with subluxation or **dislocation** (L3–L4); frog posture in high lesions.
- **Knee:** flexion contracture (high lesions); valgus (low lumbar, sacral).
- **Foot: clubfoot**, vertical talus, **calcaneus** (low lumbar), cavovarus and claw toes (sacral).
- **Rotation:** femoral anteversion; tibial torsion.

3 marks Hydrocephalus: types and clinical features

Asked: W19 (1c), S20 (1f), W20 (1c) (asked 3 times)

Types: non-communicating (obstructive: aqueduct stenosis, Chiari II, tumour) and **communicating** (not absorbed: after premature bleed or meningitis); congenital or acquired; normal-pressure in older adults.

Features in infants:

- Head growing too fast (crossing centiles); **bulging anterior fontanelle**; widened sutures; prominent scalp veins.
- **"Setting-sun" eyes**; irritability, vomiting, poor feeding, drowsiness, high-pitched cry.
- Stiff legs, delayed milestones, seizures.

Older children: morning headache, vomiting, papilloedema. NPH: gait difficulty, memory loss, incontinence.

7 marks Deformities in infantile spina bifida and their physiotherapy (2 + 5)

Asked: W17 (2b)

Deformities (2)

Region	Deformity	Why
Spine	Scoliosis, kyphosis, lordosis	Abnormal vertebrae, weak trunk, tethered cord
Hip	Flexion–adduction, subluxation or dislocation	Hip flexors and adductors work without glutei
Knee	Flexion contracture; valgus	Sitting posture; weak quadriceps
Foot	Clubfoot, vertical talus, calcaneus, cavovarus	Imbalance of working and paralysed muscles

Physiotherapy (5)

- **Assess** motor and sensory level by muscle charting; record deformities and skin.
- **Positioning:** daily **prone lying** for hip extension; hips in slight abduction; avoid frog posture.
- **Stretching:** hip flexors and adductors, hamstrings, calf; gentle (thin bones).
- **Clubfoot:** serial casting with the surgeon; then AFOs or night splints.
- **Standing frame** from about 12 months for bones and hips; orthoses by level.
- **Strengthen** working muscles and the trunk and arms.
- Parent education: handling, stretches, skin checks; latex-free equipment.

7 marks Physiotherapy of (lumbar) myelomeningocele / spina bifida

Asked: W16 (2f), S17 (2b), W21 (2f), W22 (2a) (asked 4 times)

Myelomeningocele: cord and roots in an open sac; flaccid paralysis and numb skin below the level, neurogenic bladder and bowel, hydrocephalus (Chiari II) in most.

By age

- **Newborn, before closure:** prone or side lying to protect the sac; record which muscles move.
- **Infancy:** positioning, gentle ROM, parent handling; head control, prone play, rolling, sitting, arm strength.
- **Toddler:** standing frame from about 12 months; parapodium or RGO from about 2 years in high lesions.
- **School age:** gait training with orthoses and crutches; wheelchair skills; fitness; self-care.

Lumbar level: key points

- **Strengthen** working muscles (hip flexors, adductors, quadriceps, trunk) and the arms for crutches.
- **Prevent deformity:** prone lying, stretch hip flexors and adductors, hamstrings, calf; night splints.
- **Orthoses by level:** no quadriceps → RGO or HKAFO + walker; weak quadriceps → KAFO; strong quadriceps (L4) → **AFOs + crutches**; sacral → AFOs or none.
- **Gait:** standing balance → walker → crutches (swing-through or four-point); stairs; falls.
- Wheelchair for distances.

Latex-free equipment; skin checks twice a day; no heat on numb skin; refer for shunt signs or new weakness (tethered cord).

7 marks Physiotherapy management for hydrocephalus

Asked: W21 (2b)

Hydrocephalus: excess CSF in the ventricles under raised pressure; treated with a **VP shunt** or endoscopic third ventriculostomy. Child has a large, heavy head, poor head control and delayed milestones.

- **Handling:** always support the heavy head and neck when lifting and carrying; teach parents.
- **Positioning:** change head position every few hours (prevent scalp sores and flattening); avoid pressure over the shunt valve; supported side lying, prone on a wedge.
- **Development:** head control (prone on elbows, supported sitting), rolling, sitting balance, reaching, standing in a frame; follow the developmental sequence.
- **Tone and range:** stretch and position stiff legs; strengthen weak muscles.
- **Stimulation:** sensory, visual and cognitive input through play.
- **Seating** with head support.
- **Education:** shunt warning signs: vomiting, drowsiness, irritability, bulging fontanelle, fast head growth, fever, headache: seek help at once.

5B(5) Infections of the nervous system

Full notes p. 317

3 marks Kernig's and Brudzinski's signs

Asked: W19 (1d), S21 (1a) (asked 2 times)

Sign	Method	Positive
Kernig's	Supine; flex hip and knee to 90°, then try to straighten the knee	Pain and resistance in the back of the thigh stop knee extension
Brudzinski's neck	Supine; flex the neck passively, chin to chest	Hips and knees flex involuntarily
Brudzinski's contralateral leg	Flex one hip and knee passively	The other leg flexes too

Meaning: meningeal irritation (meningitis, subarachnoid haemorrhage); stretch on inflamed roots. Low sensitivity (positive in only about 5% of adults with meningitis), so a negative sign does not rule it out.

5B(6) Post-polio residual paralysis

Full notes p. 322

3 marks What is PPRP?

Asked: S24 (1c)

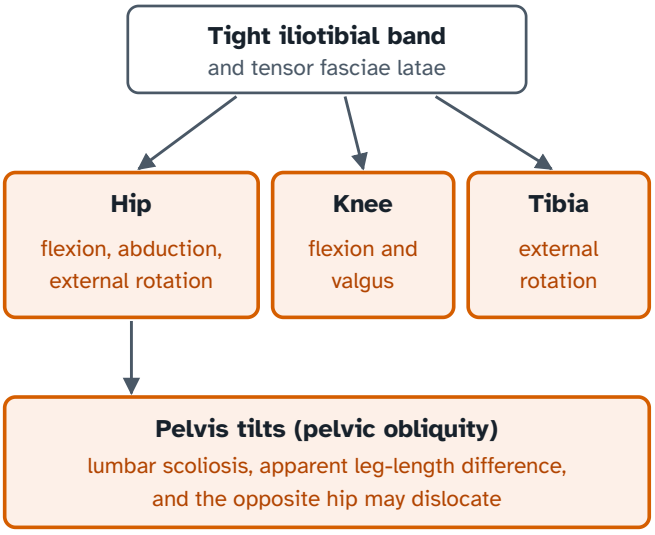
- **Post-polio residual paralysis:** the permanent weakness, wasting and deformity left once recovery from paralytic poliomyelitis has stopped, about **2 years** after the illness (residual stage).
- Poliovirus destroys **anterior horn cells**: **asymmetrical flaccid** paralysis, legs more than arms, lost reflexes, **normal sensation and bladder**.
- Muscle imbalance and growth cause deformities (hip flexion-abduction-external rotation from a tight iliotibial band, knee flexion or recurvatum, equinus), shortening and a thin, cold limb.
- Not the same as **post-polio syndrome**: new weakness and fatigue after at least 15 stable years.

7 marks Define PPRP; describe three deformities in detail

Asked: W22 (2c)

PPRP: permanent LMN paralysis, wasting and deformity remaining after recovery from paralytic polio stops (about 2 years after onset); asymmetrical, flaccid, sensation normal.

What a tight iliotibial band does



Draw this for the deformity questions. The band shortens when the child lies with hips bent, apart and turned out (the frog position) during the acute illness, then fails to grow with the bone.

▶ Text version of the flowchart

1. Hip flexion, abduction and external rotation

- **Cause:** tight **iliotibial band** and tensor fasciae latae, from lying in the frog position during the acute illness.
- **Signs:** positive Thomas and **Ober** tests; pelvic obliquity, compensatory lumbar scoliosis, apparent shortening.
- **Management:** prone lying, stretching; soft-tissue release if fixed.

2. Genu recurvatum

- **Cause:** weak quadriceps; the person locks the knee back for stability; equinus adds to it.
- **Management:** leave mild recurvatum (it stabilises the knee); >15° needs a brace (KAFO) or osteotomy.

3. Equinus (equinovarus)

- **Cause:** weak tibialis anterior with a working calf, plus gravity.
- **Signs:** foot drop, high-stepping gait, tripping.
- **Management:** calf stretching, AFO; tendon transfer (tibialis posterior) later.

Others: knee flexion, knee valgus with external tibial rotation, calcaneus foot, paralytic scoliosis, flail shoulder.

7 marks Physiotherapy management of post-polio residual paralysis

Asked: W18 (2f)

- **Assess:** dated muscle chart (MRC 0–5), contractures (Thomas, Ober, popliteal angle), limb length, gait, function.
- **Strengthen:** grade ≥3 muscles with progressive resistance (8–12 reps, 2–3 days a week, short of fatigue); grade 1–2 with assisted and gravity-eliminated work, hydrotherapy.
- **Stretch:** iliotibial band, hip flexors, hamstrings, calf; daily prone lying; serial casts or night splints.
- **Orthoses:** AFO for foot drop; **floor-reaction orthosis** (weak quadriceps, normal hip) or **KAFO** with drop-lock knee (weak hip too); shoe raise for shortening over 2 cm; crutches or stick.
- **Gait and function:** walking with orthosis, stairs, transfers, daily activities.
- **Around surgery:** strengthen the donor muscle (grade 4–5) before a tendon transfer; afterwards re-educate using the donor's old action with biofeedback; protect the transfer.
- **Education:** home programme, skin checks under orthoses, pacing, signs of post-polio syndrome.

Do not stretch away a mild recurvatum or equinus that keeps a weak knee stable.

7 marks PPRP: clinical features and physiotherapy of its complications

Asked: S25 (2e)

PPRP: residual asymmetrical flaccid paralysis after polio; thin, short, cold limb; normal sensation; gaits: hand-on-thigh (weak quadriceps), Trendelenburg, gluteus maximus lurch, high-stepping, short-limb.

Complication	Physiotherapy
Contractures and deformities	Stretching, prone lying, serial casts, night splints, orthoses; rehabilitation after release
Joint instability	Strengthen remaining muscles; floor-reaction orthosis or KAFO; keep a helpful mild recurvatum
Limb-length difference	Shoe raise, gait training, back and hip strengthening
Scoliosis, pelvic obliquity	Trunk strengthening, seating support, breathing exercises; release of hip contracture
Osteoporosis, fractures	Weight bearing in orthoses, falls prevention, gentle stretching force
Overuse pain, early arthritis	Joint protection, correct crutch use, weight control, pacing
Cold, bluish limb	Warm clothing, skin care, active movement
Post-polio syndrome	Pacing, energy conservation, moderate non-fatiguing exercise (warm water), orthosis review

5B(7) Duchenne muscular dystrophy and other myopathies

Full notes p. 334

3 marks Gowers' sign

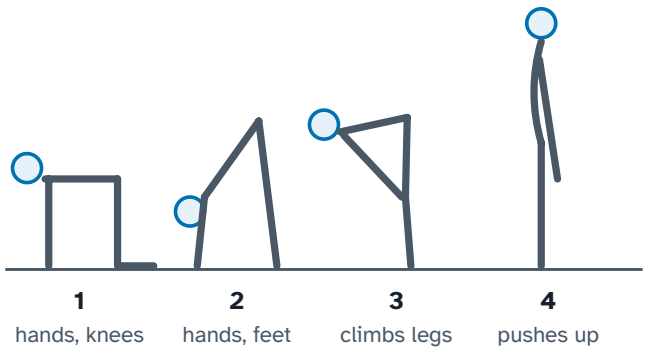
Asked: S16 (1e), S24 (1e) (asked 2 times)

The way a child with **weak hip and knee extensors** rises from the floor:

1. Turns face down and gets onto hands and knees.
2. Hands and feet with the hips high.
3. **Walks the hands up** the knees and thighs ("climbing up his own legs").
4. Pushes the trunk upright.

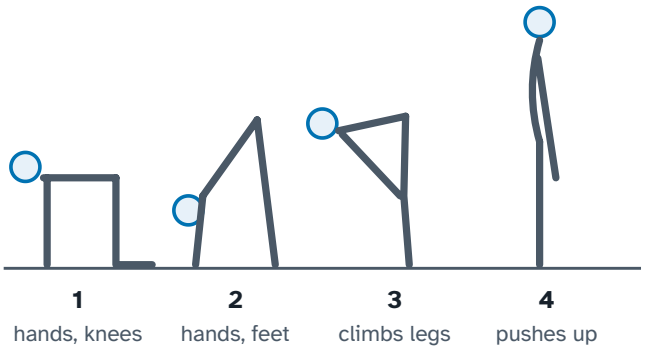
Shows **proximal lower-limb weakness**: typical of DMD, also other myopathies and SMA.

Gowers' sign: rising from the floor in four steps



The child uses the arms to "climb up" his own legs because the hip and knee extensors are too weak to lift the trunk.

Gowers' sign: rising from the floor in four steps



The child uses the arms to "climb up" his own legs because the hip and knee extensors are too weak to lift the trunk.

Gait deviations

- **Waddling (Trendelenburg) gait:** pelvis drops and trunk sways over the stance leg (weak hip abductors).
- **Increased lumbar lordosis** with shoulders held back: weak hip extensors, so the line of gravity is kept behind the hips.
- **Wide base** for balance.
- **Toe walking** (equinus): tight calves and weak dorsiflexors; no heel strike.
- **Knee locked back** in stance as quadriceps weaken.
- Slow, short steps; arms behind the trunk; frequent falls; cannot run or jump.

Calves look big but are weak (pseudohypertrophy: fat and fibrous tissue).

7 marks Gowers' sign and gait deviations in DMD

Asked: S26 (2e)

Gowers' sign

Rising from the floor by turning prone, getting onto hands and feet with hips high, then **walking the hands up the thighs** to push the trunk upright, because hip and knee extensors are weak.

7 marks Complications of DMD and their management

Asked: S18 (2f), W24 (2e) (asked 2 times)

Complication	What to know	Prevention and physiotherapy
Contractures	Ankle plantarflexors, iliotibial band, hip flexors first; later knees, elbows, wrists	Home stretching 4–6 times a week; **night AFOs**; standing; serial casting
Scoliosis	Common after walking is lost	Spine check every visit; **symmetrical seating** with lateral supports; refer early for fusion

Complication	What to know	Prevention and physiotherapy
Falls and fractures	Low-trauma fractures can end walking	Fall-proof home and school; vitamin D, calcium; get the child up quickly after a fracture
Respiratory failure	Weak cough, night hypoventilation	Lung volume recruitment (FVC ≤60%); **assisted cough** (FVC <50%, PCF <270 L/min); NIV
Cardiomyopathy	Often silent	Regular cardiac review; avoid overexertion
Weight problems	Obesity on steroids; undernutrition later	Activity, diet advice
Pain, mood, learning	Back pain, anxiety	Positioning, seating, school support, counselling

7 marks Duchenne muscular dystrophy: features, principles and physiotherapy

Asked: S16 (2f), W22 (2d), S24 (2c), S25 (2d) (asked 4 times)

DMD: X-linked recessive absence of **dystrophin**; boys; onset 2–5 years: late walking, falls, difficulty running and climbing stairs, toe walking, **calf pseudohypertrophy**, **Gowers' sign**, waddling gait, lordosis; loss of walking about 10–14 years; then scoliosis, respiratory failure, cardiomyopathy. CK very high.

Principles

- Preserve walking and range as long as possible; keep active **without overworking** fragile muscle; prevent deformity and falls; support breathing; keep independent and included.
- **Avoid eccentric and heavy resistance** exercise; stop before fatigue.

By stage

- **Early ambulatory:** home stretching 4–6 times a week; night AFOs; submaximal aerobic play (**swimming, cycling**); buggy for long distances; fall prevention; parent education.
- **Late ambulatory:** serial casting; **KAFOs** with locked knees to prolong standing and walking; manual wheelchair; standing frame.
- **Early non-ambulatory:** power wheelchair with custom seating; supported standing; arm stretching; breathing exercises; spine checks.
- **Late non-ambulatory:** seating and pressure care; lung volume recruitment and assisted cough; arm supports; hoist transfers.

Dark urine or next-day pain means the activity was too hard. Never stop steroids suddenly.

Index by paper

Each paper's short questions (Q1 at 3 marks, Q2 at 7 marks) with the page of the model answer. Sit a paper in 100 minutes (five of each), then check.

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Abbreviations

1RM: One-repetition maximum
ABI: Ankle–brachial index
ACA: Anterior cerebral artery
ADA: American Diabetes Association
AF: Atrial fibrillation
AFO: Ankle–foot orthosis
AHA: American Heart Association
ALS: Amyotrophic lateral sclerosis
APTA: American Physical Therapy Association
ASA: American Stroke Association
ASIA: American Spinal Injury Association
ATNR: Asymmetrical tonic neck reflex
BP: Blood pressure
BPPV: Benign paroxysmal positional vertigo
CFCS: Communication Function Classification System
CIMT: Constraint-induced movement therapy
CK: Creatine kinase
CMV: Controlled mandatory ventilation
CNS: Central nervous system
CP: Cerebral palsy
CPP: Cerebral perfusion pressure
CRPS: Complex regional pain syndrome
CSF: Cerebrospinal fluid
CVA: Cerebrovascular accident (stroke)
D2: Diagonal 2 (PNF pattern)
DMD: Duchenne muscular dystrophy
DVT: Deep vein thrombosis
EAU: European Association of Urology
EDSS: Expanded Disability Status Scale
EMG: Electromyography
FVC: Forced vital capacity
GBS: Guillain–Barré syndrome
GCS: Glasgow Coma Scale
GMFCS: Gross Motor Function Classification System
HKAFO: Hip–knee–ankle–foot orthosis
HR: Heart rate
HRmax: Maximum heart rate
ICF: International Classification of Functioning, Disability and Health
ICP: Intracranial pressure
ICU: Intensive care unit
IP: Interphalangeal
IQ: Intelligence quotient
ISNCSCI: International Standards for Neurological Classification of Spinal Cord Injury
IWGDF: International Working Group on the Diabetic Foot
KAFO: Knee–ankle–foot orthosis
LMN: Lower motor neuron
LSVT: Lee Silverman Voice Treatment
MAP: Mean arterial pressure
MAS: Modified Ashworth Scale
MCA: Middle cerebral artery
MCP: Metacarpophalangeal
MDS: International Parkinson and Movement Disorder Society
MIP: Maximal inspiratory pressure
MND: Motor neuron disease
MRC: Medical Research Council
MRI: Magnetic resonance imaging
MS: Multiple sclerosis
NDT: Neurodevelopmental treatment (Bobath)
NIV: Non-invasive ventilation
PCA: Posterior cerebral artery
PCF: Peak cough flow
PD: Parkinson's disease
PNF: Proprioceptive neuromuscular facilitation
PPRP: Post-polio residual paralysis
PT: Physiotherapy
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
SCI: Spinal cord injury
SMA: Spinal muscular atrophy
SMART: Specific, measurable, achievable, relevant, time-bound
TENS: Transcutaneous electrical nerve stimulation
TIA: Transient ischaemic attack
TUG: Timed Up and Go test
UMN: Upper motor neuron
VC: Vital capacity
VP: Ventriculoperitoneal (shunt)
WHO: World Health Organization